

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414591
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Hines; Douglas Paul et al.

Surgical helmet assembly with a reconfigurable chin bar

Abstract

A face frame including features for coupling the surgical garment to a surgical helmet. The face frame (524) comprises a lower beam (528) connected by a pair of posts (526, 527). The posts of the face frame comprise a face frame coupler (552, 554) configured to removably couple the face frame to a surgical helmet.

Inventors:	Hines; Douglas Paul (Kalamazoo, MI), Belikov; Sergey (Kalamazoo, MI), Jefferis; Ryan (Seattle, WA)
Applicant:	Stryker Corporation (Kalamazoo, MI)
Family ID:	1000008821108
Assignee:	Stryker Corporation (Portage, MI)
Appl. No.:	17/607121
Filed (or PCT Filed):	April 30, 2020
PCT No.:	PCT/US2020/030745
PCT Pub. No.:	WO2020/223497
PCT Pub. Date:	November 05, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220151321 A1	May. 19, 2022

Related U.S. Application Data

us-provisional-application US 62952808 20191223

Publication Classification

Int. Cl.: **A41D13/12** (20060101); **A41D13/11** (20060101); **A42B3/28** (20060101); **A62B18/04** (20060101)

U.S. Cl.:

CPC **A41D13/1218** (20130101); **A41D13/1153** (20130101); **A42B3/28** (20130101); **A62B18/04** (20130101);

Field of Classification Search

CPC: A41D (13/1218); A41D (13/1153); A62B (3/225); A62B (3/205); A62B (3/22); A62B (3/32-328); A62B (18/04); A62B (18/045); A62B (17/04)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1481019	12/1923	Luebbert	N/A	N/A
1622311	12/1926	Fischer	N/A	N/A
1973677	12/1933	Woodford	N/A	N/A
4028739	12/1976	Bell et al.	N/A	N/A
6892400	12/2004	Choi et al.	N/A	N/A
7159249	12/2006	Dennis et al.	N/A	N/A
7207072	12/2006	Halsnes et al.	N/A	N/A
7735156	12/2009	VanDerWoude et al.	N/A	N/A
7752682	12/2009	VanDerWoude et al.	N/A	N/A
8020220	12/2010	McElroy et al.	N/A	N/A
8056152	12/2010	Brace	N/A	N/A
8234722	12/2011	VanDerWoude et al.	N/A	N/A
8282235	12/2011	Gilligan	N/A	N/A
8291513	12/2011	Prinkey	N/A	N/A
8407818	12/2012	VanDerWoude et al.	N/A	N/A
8453262	12/2012	Green	N/A	N/A
8458821	12/2012	Rivas, III et al.	N/A	N/A
8819869	12/2013	VanDerWoude et al.	N/A	N/A
8899774	12/2013	Strong et al.	N/A	N/A
9173437	12/2014	VanDerWoude et al.	N/A	N/A
9400101	12/2015	Strong et al.	N/A	N/A
10253964	12/2018	Strong et al.	N/A	N/A
10420386	12/2018	Jefferis et al.	N/A	N/A
10441019	12/2018	Huh	N/A	N/A
10667568	12/2019	Rosati et al.	N/A	N/A
10750800	12/2019	Jefferis et al.	N/A	N/A
10830428	12/2019	Africa et al.	N/A	N/A
10959473	12/2020	Chen	N/A	N/A

2005/0235402	12/2004	Footte et al.	N/A	N/A
2007/0028372	12/2006	VanDerWoude	2/457	A42B 3/30
2009/0151054	12/2008	VanDerWoude	2/410	A42B 3/286
2011/0185465	12/2010	Prinkey	2/15	A42B 3/22
2014/0033406	12/2013	Lebel	2/422	A42B 3/326
2014/0075654	12/2013	McGinn	2/422	A42B 3/326
2014/0352037	12/2013	VanDerWoude et al.	N/A	N/A
2015/0082522	12/2014	Rosati et al.	N/A	N/A
2018/0084848	12/2017	Pavalarajan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203152623	12/2012	CN	N/A
106579629	12/2016	CN	N/A
210521651	12/2019	CN	N/A
211532900	12/2019	CN	N/A
2019155269	12/2018	WO	N/A

OTHER PUBLICATIONS

English language abstract and machine-assisted English translation for CN 106579629 A extracted from espacenet.com database on Dec. 16, 2021, 7 pages. cited by applicant

English language abstract and machine-assisted English translation for CN 203152623 U extracted from espacenet.com database on Dec. 16, 2021, 5 pages. cited by applicant

English language abstract and machine-assisted English translation for CN 210521651 U extracted from espacenet.com database on Dec. 16, 2021, 6 pages. cited by applicant

English language abstract and machine-assisted English translation for CN 211532900 A extracted from espacenet.com database on Dec. 16, 2021, 8 pages. cited by applicant

International Search Report for Application No. PCT/US2020/030745 dated Sep. 18, 2020, 2 pages. cited by applicant

Partial International Search Report for Application No. PCT/US2020/030745 dated Jul. 24, 2020, 2 pages. cited by applicant

Primary Examiner: Murphy; Victoria

Assistant Examiner: Toich; Sara K

Attorney, Agent or Firm: Howard & Howard Attorneys PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a National Stage of International Patent Application No. PCT/US2020/030745, filed on Apr. 30, 2020, which claims priority to and all the benefits of U.S. Provisional Patent Application No. 62/842,106, filed on May 2, 2019, and U.S. Provisional Patent Application No. 62/952,808, filed on Dec. 23, 2019, all of which are hereby expressly incorporated herein by reference in their entirety.

SUMMARY

(1) The present disclosure relates generally to a surgical helmet assembly for use with a surgical garment assembly that may be configured for attachment to the surgical helmet assembly, wherein the surgical garment assembly can be employed to provide a barrier between an individual wearing

the system and the surrounding environment.

(2) An exemplary configuration provides a surgical helmet assembly comprising a surgical garment configured for attachment to the surgical helmet. The surgical helmet may comprise a face frame including a sliding mechanism and/or a pivot to allow for reconfiguration and/or adjustment of the face frame. The face frame may also comprise a coupling feature and/or coupling member. The surgical garment includes an opening and/or an attachment member configured to removably couple the surgical garment to the surgical helmet.

(3) In another exemplary configuration, a surgical helmet assembly for use with a surgical garment including a coupling element, the surgical helmet assembly may comprise a surgical helmet. The surgical helmet may comprise a headband for securing the surgical helmet to a wearer and a ventilation unit supported by the headband. The surgical helmet assembly may further comprise a face frame that may be removably coupled to the surgical helmet. The face frame may comprise an upper beam comprising opposing regions, a first post and a second post, and a lower beam. Each of the first and second posts may comprise a first end and a second end, wherein the first end of each of the first and the second posts may be coupled to the opposing regions of the upper beam. The lower beam may extend between and connect the second ends of the first and the second posts. Each of the first post and the second post may further comprise a top member and a bottom member. The bottom member of each of the first post and the second posts may be movably coupled to the top member to provide a variable length of each of the first post and the second post to vary the distance between the upper beam and the lower beam.

(4) In yet another exemplary configuration, a surgical helmet assembly for use with a surgical garment including a coupling element, the surgical helmet assembly may comprise a surgical helmet comprising a headband for securing the surgical helmet to a wearer and a ventilation unit supported by the headband. The surgical helmet assembly may further comprise a face frame that is slidably coupled to the surgical helmet. The face frame may comprise an upper beam comprising opposing regions, a first post and a second post, and a lower beam. Each of the first post and the second post may comprise a first end and a second end, wherein the first end of each of the first and the second posts may be coupled to the opposing regions of said upper beam. The lower beam may extend between and connect the second ends of the first and the second posts. Each of the first post and the second post may further comprise a top member and a bottom member, wherein the bottom member may be movably coupled to said top member. Each of the first post and the second post may be movable between a first position defining a first configuration of the face frame and a second position defining a second configuration of the face frame such that each of the configurations of the face frame may define a different distance between the upper beam and the lower beam.

(5) In yet another exemplary configuration, a surgical helmet assembly for use with a surgical garment including a transparent shield, the surgical helmet assembly may comprise a surgical helmet comprising a coupling feature disposed on a distal portion of the surgical helmet. The coupling feature may comprise a plurality of retention features spaced about a distal surface of the coupling feature. The surgical helmet assembly may further comprise a face frame that is removably coupled to the surgical helmet. The face frame may comprise a coupling mechanism configured to operatively engage the coupling feature of said surgical helmet. The coupling mechanism and the coupling feature may further comprise complementary features to allow adjustment of the position of the face frame relative to the surgical helmet to position the transparent shield of the surgical garment at a preferred position in front of the wearer's face when disposed over the surgical helmet.

(6) In yet another exemplary configuration, a surgical helmet for use with a surgical garment including a coupling element, the surgical helmet may comprise a headband, a ventilation unit supported by the headband, and a face frame. The face frame may comprise an upper beam comprising a first region and an opposing second region, a first post and a second post, and a lower

beam. Each of the first post and the second post may comprise a first end and an opposing second end, wherein the first end of each of the first and the second posts may be coupled to the upper beam. The lower beam may extend between and connect the second ends of the first and the second posts. The opposing ends of the lower beam may be connected to the second ends of the first and the second posts by a first joint configured to allow the lower beam to rotate relative to the first and the second posts.

(7) In yet another exemplary configuration, a surgical helmet system for use with a surgical garment including a coupling element, the surgical helmet system may comprise a surgical helmet comprising a headband for securing said helmet to a wearer and a ventilation unit supported by the headband. The surgical helmet system may also comprise a first face frame configured to be removably coupled to the surgical helmet. The first face frame may comprise a first dimension, wherein the first dimension may comprise one of a first height defined by the distance between an upper beam and a lower beam, and a first width defined by the distance between a pair of posts that extend between said upper beam and said lower beam. The surgical helmet system may also comprise a second face frame configured to be removably coupled to the surgical helmet. The second face frame may comprise a second dimension, wherein the second dimension may comprise one of a second height defined by the distance between an upper beam and a lower beam, and a second width defined by the distance between a pair of posts that extend between said upper beam and said lower beam. The first dimension of the first face frame may be larger than the second dimension of the second face frame such that the first face frame is configured to fit a wearer with a larger head and the second face frame is configured to fit a wearer with a smaller head.

(8) In yet another exemplary configuration, a surgical helmet assembly may comprise a surgical helmet. The surgical helmet may comprise a headband and a ventilation unit supported by the headband. The surgical helmet assembly may further comprise a first face frame configured to be removably coupled to the surgical helmet. The first face frame may comprise a first upper beam comprising opposing regions and a first coupling feature disposed between the opposing regions. The first face frame may also comprise a first pair of posts that each comprise a first end and a second end. The first end of each of the first pair of posts may be coupled to the opposing ends of the first upper beam. The first face frame may also comprise a first lower beam extending between and connecting the second ends of the first pair of posts. The first face frame may further comprise a first attachment element at least partially disposed within the first face frame, wherein the first coupling feature and the first attachment element may define a first coupling height.

(9) The surgical helmet assembly may further comprise a second face frame configured to be removably coupled to the surgical helmet. The second face frame may comprise a second upper beam comprising opposing regions and a second coupling feature disposed between the opposing regions. The second face frame may also comprise a second pair of posts that each comprise a first end and a second end. The first end of each of the second pair of posts may be coupled to the opposing ends of the second upper beam. The second face frame may also comprise a second lower beam extending between and connecting the second ends of the second pair of posts. The second face frame may further comprise a second attachment element at least partially disposed within the second face frame, wherein the second coupling feature and the second attachment element defining a second coupling height.

(10) The surgical helmet assembly may further comprise a surgical garment. The surgical garment may comprise a shield having an upper portion and an opposing lower portion, an aperture in the upper portion of the shield, a first coupling element disposed on the lower portion of the shield, and a second coupling element disposed on the lower portion of the shield. The distance between the aperture and the first coupling element may define a third coupling height, and the distance between the aperture and the second coupling element may define a fourth coupling height. The first coupling height of the first face frame may be configured to correspond to the third coupling height of the surgical garment to allow the surgical garment to be removably coupled to the first

face frame. The second coupling height of the second face frame may be configured to correspond to the fourth coupling height of the surgical garment to allow the surgical garment to be removably coupled to the second face frame.

(11) In yet another exemplary configuration, a surgical helmet assembly comprises a surgical helmet comprising a headband for securing the helmet to a wearer and a ventilation unit supported by the headband. The surgical helmet assembly may further comprise a first face frame configured to be removably coupled to the surgical helmet. The first face frame may comprise a first upper beam comprising opposing regions and a first coupling feature disposed between the opposing regions. The first face frame may also comprise a first pair of posts that each comprise a first end and a second end. The first end of each of the first pair of posts may be coupled to the opposing ends of the first upper beam. The first face frame may also comprise a first lower beam extending between and connecting the second ends of the first pair of posts. The first face frame may further comprise a first attachment element at least partially disposed within the first face frame, wherein the first coupling feature and the first attachment element may define a first coupling height. The surgical helmet assembly may further comprise a second face frame configured to be removably coupled to the surgical helmet. The second face frame may comprise a second upper beam comprising opposing regions and a second coupling feature disposed between the opposing regions. The second face frame may also comprise a second pair of posts that each comprise a first end and a second end. The first end of each of the second pair of posts may be coupled to the opposing ends of the second upper beam. The second face frame may also comprise a second lower beam extending between and connecting the second ends of the second pair of posts. The second face frame may further comprise a second attachment element at least partially disposed within the second face frame, wherein the second coupling feature and the second attachment element defining a second coupling height;

(12) The surgical helmet assembly may further comprise a first surgical garment. The first surgical garment may comprise a first shield having an upper portion and an opposing lower portion, a first aperture in the upper portion of the first shield, and a first coupling element disposed on the lower portion of the first shield. The first aperture and the first coupling element may define a third coupling height. The surgical helmet assembly may further comprise a second surgical garment. The second surgical garment may comprise a second shield having an upper portion and an opposing lower portion, a second aperture in the upper portion of the second shield, and a second coupling element disposed on the lower portion of the second shield. The second aperture and the second coupling element may define a fourth coupling height. The first coupling height of the first face frame may be configured to correspond to the third coupling height of the first surgical garment to allow the first surgical garment to removably couple to the first face frame. The second coupling height of the second face frame may be configured to correspond to the fourth coupling height of the second surgical garment to allow the second surgical garment to removably couple to the second face frame.

(13) In yet another exemplary configuration, a surgical helmet assembly for use with a surgical garment including a transparent shield may comprise a surgical helmet. The surgical helmet may comprise a mounting feature disposed on a distal portion of the surgical helmet. The assembly may further comprise a face frame removably coupled to the surgical helmet. The face frame may comprise a first post and a second post laterally spaced from one another. Each of the first and second posts may comprise a first end and a second end. The face frame may further comprise a lower beam, a first upper bracket, and a second upper bracket. The lower beam may extend between the second end of each of the first and second posts. The first upper bracket may be coupled to and extend proximally from the first end of the first post. The second upper bracket may be coupled to and extend proximally from the first end of the second post. The face frame may further comprise a face frame coupler configured to operatively engage the mounting feature of the surgical helmet.

(14) In yet another exemplary configuration, a face frame for use with a surgical helmet including a mounting feature may comprise a first post and a second post laterally spaced from one another. Each of the first and the second posts may comprise a first end and a second end. The face frame may further comprise a lower beam and a face frame coupler. The lower beam may extend between the second end of each of the first and the second posts. The face frame coupler may be positioned proximate to the first end of each of the first and the second posts such that the face frame coupler may be configured to removably engage the mounting feature of the surgical helmet to allow for removal of the face frame from the surgical helmet to reduce the weight of the surgical helmet when the face frame is not needed.

(15) In yet another exemplary configuration, a face frame for use with a surgical helmet including a mounting feature may comprise a first post and a second post laterally spaced from one another. Each of the first and the second posts may comprise a first end and a second end. The face frame may further comprise a lower beam, a first upper bracket, and a second upper bracket. The lower beam may extend between the second end of each of the first and the second posts. The first upper bracket may be coupled to and extend proximally from the first end of the first post. The second upper bracket may be coupled to and extend proximally from the first end of the second post. The face frame may further comprise a plurality of counterweights disposed on each of the first and the second upper brackets that may be positioned proximal to the first and second posts. The plurality of counterweights may be positioned on each of the first and second upper brackets at a distance from the first and second posts sufficient to maintain sagittal position of the center of gravity of the surgical helmet when the face frame is coupled to the surgical helmet.

(16) In yet another exemplary configuration, a surgical garment for use with a surgical helmet including a coupling member may comprise a fabric defining an opening. The surgical garment may further comprise a transparent shield disposed within the opening. The transparent shield may comprise an upper portion, a lower portion, and an attachment element. The attachment element may be coupled to the upper portion of the transparent shield and configured to removably couple with the coupling member of the surgical helmet. The surgical garment may further comprise an adjustment device coupled to the lower portion of the transparent shield, the adjustment device being positionable to vary the curvature of the lower portion of the transparent shield.

(17) These and other configurations, features, and advantages of the present disclosure will be apparent to those skilled in the art. The present disclosure is not intended to be limited to or by these configurations, embodiments, features, and/or advantages.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Referring now to the drawings, exemplary illustrations are shown in detail. Although the drawings represent schematic embodiments and/or exemplary configurations, the drawings are not necessarily to scale, and certain features may be exaggerated to better illustrate and explain an innovative aspect of an exemplary configuration. Furthermore, the exemplary illustrations described herein are not intended to be exhaustive or otherwise limiting or restricting to the precise form and configuration shown in the drawings and disclosed in the following detailed description.

(2) Advantages of the present disclosure will be readily appreciated as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

(3) FIG. 1A is a perspective view of a surgical apparel system including a first configuration of a surgical helmet assembly and a surgical garment including a transparent shield disposed over the surgical helmet assembly, with the surgical helmet assembly shown in phantom.

(4) FIG. 1B is a perspective view of the first configuration of the surgical helmet assembly of the

surgical apparel system of FIG. 1A.

(5) FIG. 2A is a perspective view of a surgical apparel system including a second configuration of a surgical helmet assembly and a surgical garment including a transparent shield disposed over the surgical helmet assembly, with the surgical helmet assembly shown in phantom.

(6) FIG. 2B is a perspective view of the second configuration of the surgical helmet assembly of the surgical apparel system of FIG. 2A.

(7) FIG. 3 is a perspective view of a third configuration of a surgical helmet assembly including a first reconfigurable face frame.

(8) FIG. 4A is a perspective view of a fourth configuration of a surgical helmet assembly including a second reconfigurable face frame.

(9) FIG. 4B is a partial sectional view of a sliding mechanism of the second reconfigurable face frame of FIG. 4A.

(10) FIG. 5A is a perspective view of a fifth configuration of a surgical helmet assembly including a third reconfigurable face frame.

(11) FIG. 5B is a partial sectional view of a sliding mechanism of the third reconfigurable face frame of FIG. 5A.

(12) FIG. 5C is a perspective view of the third reconfigurable face frame of FIG. 5A including a repositionable coupling figure.

(13) FIG. 5D is a zoomed perspective view of the repositionable coupling figure of FIG. 5C.

(14) FIG. 6A is a perspective view of a first configuration of a transparent shield of a surgical garment for use with a surgical helmet assembly, the transparent shield comprising an aperture and an attachment element that define a first distance.

(15) FIG. 6B is a schematic view of the transparent shield of FIG. 6A being coupled to the third reconfigurable face frame of FIG. 5A when the sliding mechanism is in a first position.

(16) FIG. 7A is a perspective view of a second configuration of a transparent shield of a surgical garment for use with a surgical helmet assembly, the transparent shield comprising an aperture and an attachment element that define a second distance.

(17) FIG. 7B is a schematic view of the transparent shield of FIG. 7A being coupled to the third reconfigurable face frame of FIG. 5A when the sliding mechanism is in an alternative position.

(18) FIG. 8 is a perspective view of a third configuration of a transparent shield of a surgical garment for use with a surgical helmet assembly, the transparent face shield comprising an aperture and a plurality of attachment elements, each of the attachment elements defining a distance from the aperture that corresponds to a position of the sliding mechanism of the third reconfigurable face frame of FIG. 5A.

(19) FIG. 9A is a perspective view of a sixth configuration of a surgical helmet assembly including a fourth reconfigurable face frame.

(20) FIG. 9B is a partial exploded view of a chin bar attachment mechanism of the fourth reconfigurable face frame of FIG. 9A.

(21) FIG. 10A is a perspective view of a seventh configuration of a surgical helmet assembly including a fifth reconfigurable face frame.

(22) FIG. 10B is a partial exploded view of a chin bar attachment mechanism of the fifth reconfigurable face frame of FIG. 10A.

(23) FIG. 11A is a perspective view of an eighth configuration of a surgical helmet assembly including a sixth reconfigurable face frame.

(24) FIG. 11B is a partial exploded view of a chin bar attachment mechanism of the sixth reconfigurable face frame of FIG. 11A.

(25) FIG. 12A is a perspective view of a ninth configuration of a surgical helmet assembly including a seventh reconfigurable face frame.

(26) FIG. 12B is a partial exploded view of the ninth configuration of the surgical helmet assembly of FIG. 12A.

- (27) FIG. 12C is a perspective view of the seventh reconfigurable face frame of the surgical helmet assembly of FIG. 12A.
- (28) FIG. 12D is a perspective view of the seventh reconfigurable face frame of the surgical helmet assembly of FIG. 12A including a repositionable mass for use with a sliding coupling feature of the surgical helmet.
- (29) FIG. 12E is a perspective view of the seventh reconfigurable face frame of the surgical helmet assembly of FIG. 12A including a repositionable mass for use with a plurality of coupling features mounted on the surgical helmet.
- (30) FIG. 13A is a perspective view of a fourth configuration of a transparent shield of a surgical garment for use with a surgical helmet assembly, the transparent shield comprising a curvature mechanism for manipulating the curvature of the transparent shield.
- (31) FIG. 13B is a bottom view of the fourth configuration of the transparent shield of FIG. 13A, the curvature mechanism in a first position.
- (32) FIG. 13C is a bottom view of the fourth configuration of the transparent shield of FIG. 13A, the curvature mechanism in a second position.
- (33) FIG. 13D is a schematic view of the transparent shield of FIG. 13A being coupled to a face frame of a surgical helmet assembly.

DETAILED DESCRIPTION

- (34) Maintaining a reliable barrier between a healthcare provider and a patient to prevent the exchange and/or transfer of particles or foreign material during a medical procedure or examination is of the utmost importance. During medical and surgical procedures, a healthcare provider may wear an assembly known as a surgical apparel system, such as the surgical apparel system **10** illustrated in FIG. 1A. An exemplary configuration of a surgical apparel system is disclosed in U.S. application Ser. No. 16/223,523, filed on Dec. 18, 2018, the contents of which are herein incorporated in their entirety.
- (35) Accordingly, the surgical apparel system **10** may comprise a surgical garment assembly comprising a surgical garment **12** configured for attachment to a surgical helmet **20**. The surgical garment **12** is configured to provide a barrier, such as a microbial barrier, between the wearer and the surrounding environment. The barrier created by the surgical garment **12** may benefit both the wearer and the patient. The barrier provided by the surgical garment **12** may substantially eliminate the likelihood that the wearer may come into contact with fluid or solid particles of matter from the patient that may be generated during the course of a surgical procedure. The barrier may substantially prevent the transfer of any foreign particles emitted by the wearer from being transferred to the patient during the surgical procedure.
- (36) Referring to FIG. 1A, the surgical garment **12** may include a surgical fabric **14** configured to cover the surgical helmet **20** and at least a portion of the head of the wearer. The surgical garment **12** may be configured as a hood, as illustrated in FIG. 1A. It will be understood that a hood **12** refers to a surgical garment **12** that covers the head and likely only extends a short distance below the neck when worn by the wearer. However, while not illustrated in the figures, it is further contemplated that the surgical garment **12** may be configured as a toga, a shirt, or a jacket. It will be understood that a toga **12** refers to a surgical garment **12** that covers the head in the same manner as a hood and extends to at least the waist when worn by the wearer.
- (37) The surgical garment **12** may be manufactured from any suitable surgical fabric **14** or combinations of fabrics to help repel and/or absorb water, debris and other contaminants. The surgical fabric **14** may include multiple layers. One such layer may be a microporous film that allows gas to pass through the fabric while still maintaining the microbial barrier. In certain configurations, the surgical fabric **14** is one that satisfies the ASTM F1670-98 standard for blood penetration resistance and/or the ASTM F1671-97B standard for viral penetration resistance. In one non-limiting example of the surgical fabric **14**, the surgical fabric **14** of the surgical garment **12** has a pore size in the approximate range of 0.05 to 0.20 microns. However, other pore sizes for the

surgical fabric **14** are also contemplated.

(38) It is further contemplated that the surgical garment **12** may be constructed of multiple different fabrics coupled to one another to define the barrier. For example, the surgical garment **12** may be primarily constructed from a barrier surgical fabric **14** and a filter fabric **16**. The filter fabric **16** may be more permeable, and hence more breathable, than the barrier surgical fabric **14** described above. The filter fabric **16** may be located in an area with a reduced risk of having a microbial particle cross the barrier, such as above the wearer's head or proximate to the crown of the wearer's head, and configured to aid in the circulation of air through the barrier. The barrier surgical fabric **14** may be attached to the filter fabric **16** using any suitable means, such as adhesive, sewing, welding, or a combination thereof.

(39) As illustrated in FIG. **1A**, the surgical garment **12** may further comprise a face shield **18**. The face shield **18** portion of the surgical garment **12** allows the wearer to see through the barrier provided by the surgical garment **12**. The face shield **18** is generally a sheet-like structure and may have a thickness of approximately 1 mm or less. The face shield **18** may be mounted and/or attached to an opening or cut-out formed in the surgical fabric **14** of the surgical garment **12**. The surgical fabric **14** may be attached around the periphery or edge of the face shield **18** by sewing, snaps, hook and loop, adhesive, welding, or combinations thereof. The face shield **18** may be constructed from a transparent material, such as a polycarbonate. One such polycarbonate is sold under the trademark LEXAN™ by Sabic. The face shield **18** of the surgical garment **12** may also be tinted to protect the wearer's eyes from heightened exposure to bright lights. Furthermore, the face shield **18** may be flexible such that the face shield **18** may be curved to accommodate different head sizes, as will be described below.

(40) The face shield **18** may further comprise an opening **49** proximate to the top portion of the face shield **18**. The opening **49** may be generally rectangular-shaped. While not illustrated in the figures, it is further contemplated that the opening **49** may be configured in the shape of a circle, an oval, a square, or any similar polygonal shape. The opening **49** may also be generally centered between the opposing ends of the face shield **18**, and serve as an alignment element and/or centering feature. Furthermore, the opening **49** may be positioned on the face shield **18** above the point of attachment for the surgical fabric **14** to the face shield **18**, so as to ensure the surgical fabric **14** covers the opening **49** to maintain the barrier provided by the surgical garment **12** between the wearer and the environment. For example, as illustrated in FIGS. **1A** and **1B**, the surgical fabric **14** of the surgical garment **12** is attached to the top of the face shield **18** at a location below the opening **49** of the face shield **18**.

(41) The surgical garment **12** may also include one or more attachment elements **58** positioned about the surgical garment **12**. The attachment elements **58** may also be referred to as a garment fastener or a second member. The attachment elements **58** are configured to releasably secure the surgical garment **12** to the surgical helmet **20**. The attachment elements **58** may take any suitable form, and may comprise metal tacks, rivets, buttons, magnets, hook and loop, snaps, or similar types of fasteners, alone or in combination. As illustrated in FIGS. **1A** and **1B**, the attachment elements **58** may be mounted to the face shield **18** of the surgical garment **12** so as to extend inwardly from the wearer side of the face shield **18**. While not illustrated in the figures, it is also contemplated that the attachment elements **58** may be positioned at any other position or location about the surgical garment **12**, including being mounted to the barrier surgical fabric **14** and/or the filtration fabric **16**. The attachment elements **58** may be mounted to the face shield **18** and/or fabric(s) **14/16** via an adhesive, rivet, snap, similar mounting device, or combination thereof.

(42) Referring again to FIG. **1A**, an example configuration of the surgical apparel system **10** is described in detail. The system **10** may include a surgical garment **12** and a surgical helmet **20**. The surgical garment **12** may be configured as a hood or a toga to be placed over the surgical helmet **20**. In the hood configuration illustrated in FIGS. **1A** and **1B**, the surgical garment **12** may be positioned over the surgical helmet **20** and configured to encompass the surgical helmet **20** and,

correspondingly, the head of the person wearing the surgical apparel system **10**, thereby covering the wearer's face and back of the head. Alternatively, if the surgical garment **12** were configured as a toga, the toga may be positioned over the surgical helmet **20** and configured to encompass the surgical helmet **20** and, correspondingly, the head, arms, shoulders, and torso of the person wearing the surgical apparel system **10**. To place the surgical garment **12** over the surgical helmet **20**, the surgical garment **12** will typically be turned inside out as the face shield **18** is aligned and affixed to the surgical helmet **20** in the manner described below. Once the face shield **18** is positioned relative to the surgical helmet **20**, the remainder of the surgical fabric **14** will typically be pulled over the wearer's head to cover the exposed components of the surgical helmet **20** and the wearer's head.

(43) Referring to FIG. **1B**, an example configuration of the surgical helmet **20** that may be utilized as part of the surgical apparel system **10** is illustrated. The surgical helmet **20** in FIGS. **2A** and **2B** includes a headband **22**. The headband **22** may be configured to encircle the wearer's head and support the surgical helmet **20**. The headband **22** may be constructed from a generally flexible or pliable material, allowing the headband **22** to conform to the general shape of the wearer's head. The headband **22** may comprise a headband control assembly **38** configured to adjust the size/shape of the headband **22**. The headband control assembly **38** may comprise a control member **39** that is manipulatable by the wearer to adjust the size of the headband **22**. For example, as illustrated in FIGS. **2A** and **2B**, the control member **39** may comprise a rotatable knob or lever. When the wearer rotates the control member **39** in one direction, the headband control assembly **38** may be configured to reduce the size, i.e., the circumference, of the headband **22**. Alternatively, when the wearer rotates the control member **39** in the opposite direction, the headband control assembly **38** may be configured to increase the size, i.e., the circumference, of the headband **22**. This allows for the headband **22** of the surgical helmet **20** to be adjusted and/or customized to securely fit on a particular individual's head irrespective of the individual's head size and/or shape.

(44) The surgical helmet **20** further includes a shell **32** that is supported by and located above the headband **22**. The shell **32** may be configured in an arcuate shape to fit over the head of the individual wearing the surgical apparel system **10**. Other helmet designs are contemplated. Many portions of the shell **32** may be formed to define voids, or open interior spaces. For example, the shell **32** may comprise a center void. The center void may be located toward the rear of the shell **32**. There may be an intake opening or aperture in the top portion of the shell **32** to provide access to the center void. The shell **32** may also include additional voids, such as a front void proximate to the front of the shell **32** and a rear void proximate to the rear of the shell **32**. The additional voids may be configured to form duct-like structures or passageways within the shell **32**. The additional voids may even be interconnected to the center void.

(45) The surgical helmet **20** may include one or more electrically-powered peripheral devices **30**, including but not limited to a ventilation assembly, a light, a camera, a microphone or other communication device, a cooling device, or combinations thereof. These devices may be mounted to and/or attached at various locations and orientations relative to the surgical helmet **20**. Each of the peripheral devices **30** may be configured to receive commands that affect the operating state of the corresponding peripheral device **30**. For example, each of the peripheral devices **30** can receive on/off commands. Alternatively, the peripheral devices **30** may receive commands that change one or more settings of the peripheral devices **30**. Such configurations allow the wearer of the surgical helmet **20** to control the operating state of the various peripheral devices **30** during the surgical procedure. In one specific example, when the peripheral device **30** is a ventilation assembly, the ventilation assembly may be configured to receive various commands to control the actuation and/or adjust the speed of the fan in the ventilation assembly. Alternatively, when the peripheral device **30** is a cooling device, the cooling device may be configured to receive commands to control the intensity of the cooling output provided by the cooling strip. When the peripheral device **30** is a microphone, the microphone may be configured to receive commands to control the volume of the audible signal produced by the microphone. When the peripheral device **30** is a light, the

light may be configured to receive commands to control the direction and/or intensity of light emitted. The peripheral devices **30** may of course be configured to be responsive to other types of commands that control the operation of the peripheral devices **30**.

(46) Wearing the surgical apparel system **10**, including the surgical garment **12**, over a wearer's head can inevitably result in the buildup of carbon dioxide and increased temperatures within the surgical garment **12** as a result of the wearer's normal breathing. An increase in temperature underneath the surgical garment **12** can also result in the buildup of water vapor on the wearer and/or the face shield **18**, resulting in the wearer's view being obstructed. In order to prevent these undesirable effects, the surgical helmet **20** of the surgical apparel system **10** may be configured for the attachment and/or inclusion of one or more peripheral devices **30** described above, such as the ventilation assembly, the cooling device, etc. Certain features of the surgical helmet **20**, the peripheral devices **30**, and the surgical garment **12** may be found in one or more of the following U.S. Patents, which are hereby incorporated by reference: U.S. Pat. Nos. 1,481,019; 1,622,311; 1,973,677; 7,735,156; 7,752,682; 8,234,722; 8,282,234; 8,407,818; 8,819,869; and 9,173,437.

(47) The ventilation assembly illustrated in FIG. **1B** is one example of a peripheral device **30** that may be incorporated into the surgical helmet **20** of the surgical apparel system **10**. While the ventilation assembly is shown as an integral component of the surgical helmet **20**, it should be appreciated that each of the other peripheral devices **30** described above may be either an integral component of the surgical helmet **20**, or may be removably coupled to the surgical helmet **20**. The surgical helmet **20** illustrated in FIG. **1B** comprises the ventilation assembly positioned within the center void of the shell **32**. The ventilation assembly may include a fan blade, impeller, propeller, fan wheel, or similar blade mechanism configured to induce air movement. The blade may be coupled to a motor configured to rotate the blade when energized by a power source. When the blade is actuated, the ventilation assembly is configured to draw air into the center void of the shell **32** through the intake opening in the top of the shell **32**. The additional voids of the shell **32** may be connected to the center void and serve as ducts for dispersing the air drawn into the center void.

(48) The exemplary ventilation assembly **30** may include a front bellows **36** or duct-like structure that extends forward from the front void in the front of the shell **32** and connects to a front nozzle **40**. The front nozzle **40** may be mounted to the front of the headband **22**. The ventilation assembly **30** may further include a rear bellows **34** or duct-like structure that extends from the rear void in the rear of the shell **32** to a rear nozzle (not shown in figures). The rear nozzle may be mounted to the back of the headband **22**. When the ventilation assembly **30** of the surgical helmet **20** is actuated, the fan draws air in through the surgical garment **12** into the opening in the top of the shell **32** and disperses the air outward through the additional voids. For example, the ventilation assembly **30** may be configured to draw air through the filter fabric **16** of the surgical garment **12**. The air is then discharged through front bellows **36** and rear bellows **34**, respectively. The air that flows through the front bellows **36** is discharged through the front nozzle **40** in front of the face of the wearer. The air discharged through the front nozzle **40** may be discharged against the face shield **18** and/or on the face of the wearer. The air that flows through the rear bellows **34** is discharged through the rear nozzle. The rear nozzle is positioned so as to open below the headband **22**. The air discharged from the rear nozzle can be discharged against the back of the neck of the wearer.

(49) The front nozzle **40** of the surgical helmet **20** may include a block **42**. The block **42** is the portion of the front nozzle **40** that is mounted to the headband **22** or a component of the surgical helmet **20** integral with the headband **22**. In the illustrated version of the system **10**, the block **42** is mounted to a strap that is part of the headband **22**.

(50) The surgical helmet **20** may include a face frame **24** that is coupled to the front nozzle **40** or block **42** of the surgical helmet **20**. The face frame **24** may comprise an upper beam **29** with opposing lateral portions **29A**, **29B**. The upper beam **29** may be positioned to generally span across the forehead of the wearer of the surgical helmet **20**. The upper beam **29** may also comprise coupling feature **146**, such as a protrusion **46**. The protrusion **46** may be configured to extend

distally outward and upward from the upper beam **29**. The protrusion **46** may be positioned proximate the center of the upper beam **29**. While the exemplary configuration of the upper beam **29** illustrated in FIGS. **1A** and **1B** only includes a single protrusion **46**, it is further contemplated that the upper beam **29** may comprise multiple protrusions **46** spaced along the length of the upper beam **29**.

(51) The face frame **24** may comprise a first post **26** and a second post **27**. Each of the first post **26** and second post **27** may further comprise an upper portion **26A**, **27A**, and a lower portion **26B**, **27B**, respectively. The upper portion **26A**, **27A** of each of the first post **26** and second post **27** may be coupled to the opposing lateral portions **29A**, **29B** of the upper beam of the face frame **24** and extend in a generally downward direction from the upper beam **29**. The first post **26** and second post **27** of the face frame **24** may be constructed from a generally flexible or pliable material.

(52) A lower beam **28** may extend between the opposed lower portions **26B**, **27B** of the first and second posts **26**, **27**. The face frame **24** is formed so that the lower beam **28** is located below and slightly forward of the chin of the person wearing the surgical helmet **20**. The lower beam **28** may be bowed outwardly from the opposed lower portions **26B**, **27B** of the posts **26**, **27**. The face frame **24** may extend outwardly from the front nozzle **40** and/or block **42** of the surgical helmet **20** such that the face frame **24** is positioned forward of and generally encircles the face of the wearer when the surgical helmet **20** is secured to the wearer's head.

(53) A plurality of coupling members **48**, such as magnets, hook and loop, metal rivets, snaps, or similar type fasteners, may be mounted to the face frame **24** and configured to align and/or attach to the face shield **18** of the surgical garment **12**. Each coupling member **48** may be positioned on the face frame **24** proximate to the opposed free ends of the first and second posts **26A**, **26B** and/or adjacent opposing ends of the lower beam **28**. Alternatively, the coupling members **48** of the surgical helmet **20** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements **58** of the face shield **18**, as described above, to releasably secure the surgical garment **12** to the surgical helmet **20**. For example, the protrusion **46** extending from the front nozzle **40** may be replaced with a coupling member **48** configured to cooperate with a complementary attachment element **58** on the surgical garment **12**.

(54) As described above, in one configuration of the system **10**, the face shield **18** of the surgical garment **12** may comprise an opening **49** proximate the top edge of the face shield **18**. The opening **49** in the face shield **18** may be configured to receive the protrusion **46** protruding from the front nozzle **40** of the surgical helmet **20** or from another portion of the surgical helmet. The opening **49** and the protrusion **46** may be configured to releasably secure the face shield **18** and/or surgical garment **12** to the surgical helmet **20**. Furthermore, the opening **49** and the protrusion **46** may serve as an alignment feature configured to align the face shield **18** with the surgical helmet **20**, such that the face shield **18** will be positioned in front of the wearer's face when the system **10** is worn.

While not shown in the figures, it should be understood that it has been contemplated that the face shield **18** may include additional openings **49** spaced laterally across the upper portion of the face shield **18** to provide points of attachment between the face shield **18** and surgical helmet **20**, and the surgical helmet **20** may be configured to include additional protrusions **46** correspondingly arranged relative to the additional openings **49** of the face shield **18**. For example, a plurality of protrusions **46** may extend from the headband **22** and/or front nozzle **40**, and the face shield **18** may be configured to include complementary openings **49** laterally spaced across the upper portion of the face shield **18** that releasably engage the plurality of protrusions **46** when attaching the surgical garment **12** to the surgical helmet **20**.

(55) Furthermore, as described above, the face shield **18** and/or surgical fabric **14** may comprise a plurality of attachment elements **58** arranged about the surgical garment **12**. In the example configuration of the surgical garment **12** that is illustrated in FIG. **1A**, the attachment elements **58** of the surgical garment **12** may be arranged and/or positioned on the face shield **18** so that, when the protrusion **46** of the surgical helmet **20** is seated in the opening **49** of the face shield **18** and the

face shield **18** is flexed around the face frame **24**, each of the garment attachment elements **58** will abut and latch to a complementary magnet or other suitable coupling member **48** on the surgical helmet **20**. Referring to the example configuration of the system **10** illustrated in FIG. **1A**, the surgical garment **12** comprises the opening **49** proximate to the top portion of the face shield **18** and a plurality of attachment elements **58** are positioned about the periphery of the face shield **18**. The attachment elements **58** may be spaced about the periphery of the face shield **18** to matingly engage complementary coupling members **48**, such as magnets, on the face frame **24** of the surgical helmet **20**. While the surgical garment **12** illustrated in the figures comprises the opening **49**, it is also contemplated that the opening **49** is not required in certain configurations. Alternatively, the attachment elements **58** may also be configured to couple and/or align the face shield **18** with the surgical helmet **20**.

(56) In operation, once the opening **49** in the face shield **18** is seated on the protrusion **46** of the surgical helmet **20**, the face shield **18** may then be flexed around the surgical helmet **20** and/or face frame **24** to matingly engage the attachment elements **58** spaced about the periphery of the face shield **18** with the complementary coupling members **48** on the face frame **24** of the surgical helmet **20**. The size of the face shield **18**, as well as the spacing and/or position of the attachment elements **58** on the surgical garment **12** may be changed to alter the curvature and/or shape of the face shield **18** when attached to the surgical helmet **20**. For example, the attachment elements **58** on the surgical garment **12** may be spaced closer together to reduce the curvature of the face shield **18** when it is attached to the surgical helmet **20**. Alternatively, the attachment elements **58** on the surgical garment **12** may be spaced farther apart to increase the curvature of the face shield **18** when it is attached to the surgical helmet **20**. Furthermore, the curvature and/or arc of the face shield **18** may also be modified or adapted by the size and/or shape of the attachment element(s) **58** of the face shield **18** and/or the corresponding coupling member(s) **48** of the surgical helmet **20**. The size of the protrusion **46** may be similarly adapted to manipulate the curvature or arc of the face shield **18**. Altering the curvature of the face shield **18** may help to reduce glare or provide an expanded/reduced peripheral view through the face shield **18**. While not illustrated in the figures, it should be understood that alternative configurations for securing the surgical garment **12** and/or face shield **18** to the surgical helmet **20** are also contemplated. For example, in one alternative configuration, the transparent face shield **18** may not include a rectangular opening **49**, but instead comprise only a plurality of attachment elements **58**, such as rivets or magnets, spaced about the transparent face shield **18** and/or surgical garment **12** and configured to couple to complementary magnets or similar coupling members **48** spaced about the face frame **24** of the surgical helmet **20**. For example, the complementary magnets or similar coupling members **48** may be secured to the shell **32**, headband **22**, and/or face frame **24**. The surgical garment **12** and the surgical helmet **20** of the surgical apparel system **10** described above are typically removably coupled to allow for disposal of the surgical garment **12** and reuse of the surgical helmet **20** following a procedure or exam.

(57) Referring to FIGS. **2A** and **2B**, an alternative configuration of a surgical apparel system **110** is illustrated. It should be appreciated that the various configurations of the surgical apparel system **110** may include similar elements that may be identified by reference numerals that are incremented by 100. It should be understood that those elements including reference numerals which are incremented by 100 can have the same features as described above. The surgical apparel system **110** may comprise a surgical garment assembly comprising a surgical garment **112** configured for attachment to a second configuration of a surgical helmet **120**. As described above, the surgical garment **112** may be configured to provide a barrier, such as a microbial barrier, between the wearer and the surrounding environment. The barrier created by the surgical garment **112** may benefit both the wearer and the patient. The barrier provided by the surgical garment **112** may substantially eliminate the likelihood that the wearer may come into contact with fluid or solid particles of matter from the patient that may be generated during the course of a surgical procedure.

The barrier may substantially prevent the transfer of any foreign particles emitted by the wearer from being transferred to the patient during the surgical procedure.

(58) The second configuration of the surgical apparel system **110** may comprise a surgical garment assembly comprising a surgical garment **112** configured for attachment to a surgical helmet **120**. As described above, the surgical garment **112** may provide a barrier, such as a microbial barrier, between the wearer and the surrounding environment. The barrier created by the surgical garment **112** may benefit both the wearer and the patient. The barrier provided by the surgical garment **112** may substantially eliminate the likelihood that the wearer may come into contact with fluid or solid particles of matter from the patient that may be generated during the course of a surgical procedure. The barrier may substantially prevent the transfer of any foreign particles emitted by the wearer from being transferred to the patient during the surgical procedure.

(59) Referring to FIG. 2A, the surgical garment **112** may include a fabric **114** configured to cover the surgical helmet **120** and at least a portion of the head of the wearer. The surgical garment **112** may be configured as a hood, toga, or other similar medical garment, similar to any of the configurations of the surgical apparel system **10** described above. The surgical garment **112** may further comprise a transparent shield **118**, also referred to as a transparent face shield, and one or more attachment elements **158** positioned about the surgical garment **112**. The attachment elements **158** may also be referred to as a second member or garment fastener. The attachment elements **158** may serve as an alignment element configured to removably couple the surgical garment **112** to the surgical helmet **120**. Furthermore, the attachment elements **158** may be positioned proximate the outer perimeter of the transparent shield **118** such that the fabric **114** covers the attachment elements **158**. This may serve to ensure the fabric **114** covers the attachment elements **158** to maintain the barrier provided by the surgical garment **112** between the wearer and the environment.

(60) The attachment elements **158** may comprise a ferromagnetic material. In other words, the attachment element may comprise iron, nickel, cobalt, gadolinium, dysprosium, or alloys thereof, or combinations thereof. It is contemplated that the entirety of the attachment element **158** consists of the ferromagnetic material. It is also contemplated that the attachment element **158** comprises both ferromagnetic material and diamagnetic material. For example, the attachment elements **158** may comprise a diamagnetic material which has been coated with a ferromagnetic material. Alternatively, the attachment elements **158** may be formed from a ferromagnetic material as a core, and then coated with a plastic or similar non-magnetic coating configured to provide a sterile and/or wear-resistant surface. Other arrangements of the diamagnetic and magnetic material are contemplated for the attachment element **158**. It should be appreciated that the surgical garment **112**, and all components thereof, may be configured similarly and/or comprise the features of the surgical garment(s) **12** described above.

(61) The surgical garment **112** may also comprise a tab **147**. The tab **147** may be disposed on the wearer side or interior of the surgical garment **112**. The tab **147** may comprise a pair of opposing edges and define an opening **149**. As illustrated in FIG. 2A, the tab **147** may be formed as a portion of the transparent shield **118**. The tab **147** may define at least a portion of the opening **149**, with a portion of the opening **149** also being defined by the transparent shield **118**. It is also contemplated that the opening **149** may be entirely formed or defined within the tab **147**.

(62) Alternatively, while not illustrate in the figures, it is further contemplated that the tab **147** may be formed separate from the transparent shield **118**. In this configuration, the tab **147** may be coupled directly to the interior surface of the fabric **114** of the surgical garment **112**. The tab **147** may be formed from a plastic similar to the transparent shield **118** and may be coupled to the surgical garment **112** by an epoxy, glue, or similar adhesive. Alternatively, the tab **147** may be formed from a fabric similar to the fabric **114** of the surgical garment **112**, wherein the tab **147** may be sewn or coupled to the surgical garment **112** by an adhesive. Furthermore, the opening **149** may be entirely defined by the tab **147**.

(63) Similar to the system **10** described above, the configuration of the system **110** illustrated in

FIGS. 2A and 2B may comprise one or more peripheral devices **130**, such as a ventilation assembly. The ventilation assembly **130** illustrated in FIG. 2B, is one example of a peripheral device **130** that may be incorporated into the surgical helmet **120** of the surgical apparel system **110**. While the ventilation assembly **130** is shown as an integral component of the surgical helmet **120**, it should be appreciated that each of the other peripheral devices **130** described above may be either an integral component of the surgical helmet **120**, or may be removably coupled to the surgical helmet **120**. The surgical helmet **120** illustrated in FIG. 2B comprises the ventilation assembly **130** positioned within the void of the shell **132**. The ventilation assembly **130** may include a fan blade, impeller, propeller, fan wheel, or similar blade mechanism configured to induce air movement. The blade may be coupled to a motor configured to rotate the blade when energized by a power source. When the blade is actuated, the ventilation assembly **130** is configured to draw air into the void of the shell **132** through the intake opening in the top of the shell **132**. The additional voids of the shell **132** may be connected to the void and serve as ducts for dispersing the air drawn into the void to the wearer.

(64) Referring to FIG. 2B, the surgical helmet **120** may comprise a face frame **124** that extends downwardly from the front portion of the surgical helmet **120**. The face frame **124** may comprise an upper beam **129** with opposing lateral portions **129A**, **129B**. The upper beam may be positioned forward of the shell **132** of the surgical helmet **120** and configured to extend across the front of the surgical helmet **120**. The upper beam **129** may further comprise an alignment channel **145** that is defined by a pair of laterally spaced-apart sidewalls and a proximal surface that is positioned proximally from the distal surface of the upper beam **129**. The alignment channel **145** is configured to receive a tab **147** disposed on the interior of the surgical garment **112** to align and/or orient the surgical garment **112** relative to the surgical helmet **120**. As described above, the tab **147** may be integrally formed with and configured to extend from the transparent shield **118**. Alternatively, the tab **147** may be formed independent of the transparent shield **118**, wherein the tab **147** is configured to be coupled to the fabric **114** on the interior of the surgical garment **112**. However, other configurations are contemplated. The spaced-apart sidewalls of the alignment channel **145** should be spaced apart a distance greater than the width of the tab **147** to allow the tab **147** to be positioned between the spaced-apart side walls.

(65) The upper beam **129** may further comprise a coupling feature **146**, such as a protrusion, configured to removably engage the transparent shield **118** and/or surgical garment **112**. The coupling feature **146** may comprise a protrusion, magnetic member, ferromagnetic member, hook and loop, or similar coupling mechanism configured to releasably engage the opening **149** in the tab **147** to align and/or couple the surgical garment **112** to the surgical helmet **120**. For example, as illustrated in FIGS. 2A and 2B, the coupling feature **146** is realized as a protrusion **146** extending from the alignment channel **145** of the upper beam **129**. Here, the upper beam **129** comprises the alignment channel **145** described above, and the coupling feature **146** may be disposed at least partially within the alignment channel **145**. The coupling feature **146** may be positioned within the alignment channel **145** such that the top of the upper most surface of the coupling feature **146** is arranged or otherwise positioned below the top of the alignment channel **145** and/or the top surface of the upper beam **129**. The combination of the spaced-apart sidewalls of the alignment channel **145** and the coupling feature **146** may serve to align and/or orient the transparent shield **118** and/or the surgical garment **112** relative to the surgical helmet **120**. More specifically, the spaced-apart sidewalls of the alignment channel **145** may serve to guide the tab **147** such that the opening **149** in the tab **147** is directed into engagement with the coupling feature **146** as the surgical garment **112** is placed over the surgical helmet **120**.

(66) The face frame **124** may comprise a first post **126** and a second post **127**. The first and second posts **126**, **127** may be coupled to the upper beam **129**, wherein the upper beam **129** is configured to extend across the front of the surgical helmet **120**. For example, as illustrated in FIG. 2B, the first and second posts **126**, **127** may be connected to opposing ends **129A**, **129B** of the upper beam

129. The face frame **124** may be constructed from a generally flexible or pliable material.

(67) The face frame **124** may further comprise a lower beam **128** that may extend between the opposed free ends of the first and second posts **126**, **127**. The face frame **124** is formed so that the lower beam **128** is located below and slightly forward of the chin of the person wearing the surgical helmet **120**. The lower beam **128** may be bowed outwardly from the free ends of the posts **126**, **127**. The face frame **124** may extend outwardly from the upper beam **129** such that the face frame **124** is positioned forward of and generally encircles the face of the wearer when the surgical helmet **120** is secured to the wearer's head. Collectively, the combination of the upper beam **129**, the first and second posts **126**, **127**, and the lower beam **128** may also be referred to as a chin bar, as they generally define an opening positioned in front of the wearer's face and extend below the wearer's chin when the surgical helmet is positioned on top of the wearer's head.

(68) A plurality of coupling members **148** may be mounted to or within the face frame **124**. The coupling members **148** comprise magnetic material and are configured to align and/or attach the transparent shield **118** of the surgical garment **112** to the surgical helmet **120**. Each coupling member **148** may be positioned on the face frame **124** proximate to the opposed first and second posts **126**, **127** and/or adjacent opposing ends of the lower beam **128**. Alternatively, the coupling members **148** of the surgical helmet **120** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements **158** of the surgical garment **112** to releasably secure the surgical garment **112** to the surgical helmet **120**. For example, as illustrated in FIG. 2B, the coupling member **148** may be positioned on the face frame **124** at opposing ends of the lower beam **128** proximate where each of the posts **126**, **127** connects to the lower beam **128**. While the exemplary configuration of the surgical helmet **120** illustrated in FIG. 2B utilizes two coupling members **148**, it is contemplated that the surgical helmet **120** may be configured such that the face frame **124** comprises a single coupling member **148** or, in other configurations, three or more coupling members **148** may be spaced about the face frame **124**. It is contemplated that other types of coupling members **148** may be used in place of and/or in addition to those comprising magnetic materials, such as with hook and loop fasteners, snaps, coupling members comprising ferromagnetic materials, or similar type fasteners. Other configurations are contemplated.

(69) Referring to FIG. 2B, an exemplary configuration of a coupling member **148** positioned within the face frame **124** is illustrated. The coupling member **148** may comprise a distal surface. The face frame **124** may comprise a recess configured to receive the coupling feature **148**. For example, as illustrated in FIG. 2B, the coupling member **148** may be positioned within the recess of the face frame **124**, such that the distal surface of the coupling member **148** is positioned proximally to a distal surface of the face frame **124**.

(70) The coupling member **148** may comprise one of a ferromagnetic material or a magnetic material. This may include the coupling member **148** being formed of or constructed from a ferromagnetic material or a magnetic material. It is also contemplated that only a portion of the coupling member **148** comprises a ferromagnetic material or a magnetic material. For example, the coupling member **148** may be injection-molded plastic and coated with a ferromagnetic material or magnetic material. Alternatively, the coupling member **148** may be formed from a ferromagnetic material or magnetic material, and then coated with a plastic or similar coating to provide a sterile and/or wear-resistant surface. It is also contemplated that a magnet may be "over-molded" with a plastic material to define the coupling member **148**. Generally, the coupling member **148** may comprise the other of the ferromagnetic material or magnetic material relative to the attachment element(s) **158** of the surgical garment **112** in order to create a magnetic attraction between the coupling member(s) **148** and the attachment element(s) **158** to couple the surgical garment **112** to the surgical helmet **120**.

(71) Similar to the transparent face shields **18**, described above, the transparent shield **118** may comprise a portion of the surgical garment **112** that allows the wearer to see through the barrier provided by the surgical garment **112**. The transparent shield **118** is generally a sheet-like structure

and may have a thickness of approximately 1 mm or less. The transparent shield **118** may be mounted and/or attached to an opening or cut-out formed in the surgical fabric **114** of the surgical garment **112**. The surgical fabric **114** may be attached around the periphery or edge of the transparent shield **118** by sewing, snaps, hook and loop, adhesive, welding, or combinations thereof. The transparent shield **118** may be constructed from a transparent material, such as a polycarbonate. One such polycarbonate is sold under the trademark LEXAN™ by Sabic. The transparent shield **118** of the surgical garment **112** may also be tinted to protect the wearer's eyes from heightened exposure to bright lights. Furthermore, the transparent shield **118** may be flexible such that the transparent shield **118** may be curved to accommodate different head sizes, as will be described below.

(72) The transparent shield **118** may further comprise the tab **147** described above, with regard to the surgical garment **112**. The tab **147** may extend from the top portion of the transparent shield **118** and may define at least a portion of the opening **149**. The opening **149** may be generally rectangular-shaped. While not illustrated in the figures, it is further contemplated that the opening **149** may be configured in the shape of a circle, an oval, a square, or any similar polygonal shape. The opening **149** may also be generally centered between the opposing ends of the transparent shield **118** and serve as an alignment element configured to interact with the alignment channel **145** of the helmet **120** described above. Furthermore, the opening **149** may be positioned on the transparent shield **118** above the point of attachment for the surgical fabric **114** to the transparent shield **118**, so as to ensure the surgical fabric **114** covers the opening **149** to maintain the barrier provided by the surgical garment **112** between the wearer and the environment. While not illustrated in the figures, as previously discussed, the tab **147** may be formed separate from the transparent shield **118** and coupled directly to the interior of the surgical garment **112**. Aside from being formed separate from the transparent shield **118**, the tab **147** may still comprise all of the same or similar features of the tab **147** that is formed as part of the transparent shield **118**. The transparent shield **118** may further comprise one or more apertures **149** positioned in the bottom portion of the transparent shield **118** and be configured to couple the attachment elements **158** to the transparent shield **118**.

(73) Referring to FIG. 3, a third configuration of a surgical helmet **220** is illustrated. Either of the surgical garments **12**, **112** described above may be utilized with the surgical helmet **220** illustrated in FIG. 3. Furthermore, it should be appreciated that the various configurations of the surgical helmet **220** may include similar elements that may be identified by reference numerals that are incremented by 100. It should be understood that those elements including reference numerals which are incremented by 100 can have the same features as described above and/or below with regard to other configurations of the surgical helmet.

(74) Similar to the configurations of the surgical helmet **20**, **120** described above, the third configuration of the surgical helmet **220** may comprise one or more peripheral devices **230**, such as a ventilation assembly. The surgical helmet **220** illustrated in FIG. 3 comprises a ventilation assembly **230** positioned within the center void of a shell **232** of the surgical helmet **220**. The ventilation assembly **230** may include a fan blade, impeller, propeller, fan wheel, or similar blade mechanism configured to induce air movement. When the blade is actuated, the ventilation assembly **230** is configured to draw air into the center void of the shell **232** through the intake opening in the top of the shell **232**. The ventilation assembly **230** may include a front bellows **236** that extends forward from the front void in the front of the shell **232** and connects to a front nozzle **240**. The front nozzle **240** may be mounted to the front of the headband **222**. The ventilation assembly **230** may further include a rear bellows **234** that extends from the rear void in the rear of the shell **232** to a rear nozzle (not shown in the figures). The rear nozzle may be mounted to the back of the headband **222**. When the ventilation assembly **230** of the surgical helmet **220** is actuated, the fan draws air into the shell **232** and disperses the air outward through front bellows **236** and rear bellows **234** to the front **240** and rear nozzles, respectively.

(75) The front nozzle **240** of the surgical helmet **220** may include a block **242**. The block **242** is the portion of the front nozzle **240** that is mounted to the headband **222** or a component of the surgical helmet **220** integral with the headband **222**.

(76) The surgical helmet **220** may include a face frame **224** that is coupled to the front nozzle **240** or block **242** of the surgical helmet **220**. The face frame **224** may comprise an upper beam **229** with opposing lateral portions **229A**, **229B**. The upper beam **229** may be positioned to generally span across the forehead of the wearer of the surgical helmet **220**. The upper beam may also comprise a protrusion **246**. The protrusion **246** may be configured to extend distally outward and upward from the upper beam **229**. The protrusion **246** may be positioned proximate the center of the upper beam **229**. While the exemplary configuration of the upper beam **229** illustrated in FIG. **3** only includes a single protrusion **246**, it is further contemplated that the upper beam **229** may comprise multiple protrusions **246** spaced along the length of the upper beam **229**.

(77) The face frame **224** may comprise a first post **226** and a second post **227**. Each of the first post **226** and second post **227** may further comprise an upper portion **226A**, **227A**, and a lower portion **226B**, **227B**, respectively. The upper portion **226A**, **227A** of each of the first post **226** and second post **227** may be coupled to the opposing lateral portions **229A**, **229B** of the upper beam **229** of the face frame **224** and extend in a generally downward direction from the upper beam **229**. The first post **226** and second post **227** of the face frame **224** may be constructed from a generally flexible or pliable material.

(78) A lower beam **228** may extend between the opposed lower portions **226B**, **227B** of the first and second posts **226**, **227**. The face frame **224** is formed so that the lower beam **228** is located below and slightly forward of the chin of the person wearing the surgical helmet **220**. The lower beam **228** may be bowed outwardly from the opposed lower portions **226B**, **227B** of the posts **226**, **227**. The face frame **224** may extend outwardly from the front nozzle **240** and/or block **242** of the surgical helmet **220** such that the face frame **224** is positioned forward of and generally encircles the face of the wearer when the surgical helmet **220** is secured to the wearer's head.

(79) A plurality of coupling members **248**, such as magnets, hook and loop, metal rivets, snaps, or similar type fasteners, may be mounted to the face frame **224** and configured to align and/or attach one of the transparent shields **18**, **118** of the surgical garments **12**, **112** described above. Each coupling member **248** may be positioned on the face frame **224** proximate to the opposed free ends **226A**, **226B** of the posts **226**, **226** and/or adjacent opposing ends of the lower beam **228**.

Alternatively, the coupling members **248** of the surgical helmet **220** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements **58**, **158** of the transparent shield **18**, **118**, as described above, to releasably secure the surgical garment **12**, **112** to the surgical helmet **220**. For example, the protrusion **246** extending from the upper beam **229** may be replaced with a coupling member **248** configured to cooperate with a complementary attachment element **58**, **158** on the surgical garment **12**, **112**.

(80) The face frame **224** may further comprise a sliding mechanism **252**. The sliding mechanism **252** may comprise a portion of the first post **226** and second post **227**. As illustrated in FIG. **3**, the sliding mechanism **252** is part of the upper portions **226A**, **227A** of the posts **226**, **227**, and is positioned proximate where the posts **226**, **227** are coupled to the upper beam **229**. While not illustrated in FIG. **3**, it is contemplated that the sliding mechanism **252** may be positioned at any point along the posts **226**, **227**. For example, the sliding mechanism **252** may be formed as part of the lower portions **226B**, **227B** of the posts **226**, **227**, and may be positioned proximate where the posts **226**, **227** are coupled to the lower beam **228** of the face frame **224**. Alternatively, the sliding mechanism **252** may be positioned at an intermediate point along the length of the first and second posts **226**, **227**, between the upper beam **229** and the lower beam **228** of the face frame **224**. It is contemplated that the sliding mechanism **252** may be configured and/or utilized to removably couple the upper portions **226A**, **227A** of the posts **226**, **227** to the lower portions **226B**, **227B** of the posts **226**, **227**. It is also contemplated that the sliding mechanism **252** may be configured

and/or utilized to removably couple the posts **226**, **227** to the upper beam **229**.

(81) The sliding mechanism **252** may comprise an inner member **253** and an outer member **255** that slidably engage one another to alter and/or manipulate the length of the first post **226** and second post **227**. For example, the sliding mechanism **252** may slide between a first position, a second position, or any intermediary position between the first and second position. When the sliding mechanism **252** is in the first position, the first post **226** and second post **227** may have a first length defining a first distance between the upper beam **229** and the lower beam **228**. This may also be referred to as a first configuration of the face frame **224**. Alternatively, when the sliding mechanism **252** is in the second position, the first post **226** and second post **227** may have a second length defining a second distance between the upper beam **229** and the lower beam **228**. This may also be referred to as a second configuration of the face frame **224**. The face frame **224** may also comprise a plurality of intermediate configurations wherein the sliding mechanism **252** is positioned at a point between the first position and the second position. This allows for the length of the face frame **224** of the surgical helmet **220** to be adjusted based on the specific size and characteristics of the individual wearing the surgical helmet. For example, an individual with a small head and/or a shorter neck may desire a shorter distance between the upper beam **229** and lower beam **228** to prevent the lower beam **228** from contacting and/or becoming ensnared on the wearer's chest. Alternatively, an individual with a larger head and/or a longer neck may desire a longer distance between the upper beam **229** and lower beam **228** to maximize viewing area provided by the transparent shield **18**, **118** of the surgical garment **12**, **112** that is coupled to the surgical helmet **220**.

(82) The inner member **253** and the outer member **255** of the sliding mechanism **252** may also comprise complementary features such as a recess or aperture **254** and protrusion **256** to define the various lengths of the first post **226** and second post **227**. For example, the outer member **255** may comprise a plurality of apertures **254** spaced along the length of the inner member **253**. The inner member **253** may then comprise at least one protrusion **256** configured to matingly engage the plurality of apertures **254** as the outer member **255** is slid along the length of the inner member **253**. The protrusion **256** when engaging one of the plurality of recess **254** may be configured to create a friction fit holding the inner member **253** and outer member **255** in place relative to one another until an external force is applied to move the protrusion **256** from the current recess **254** to one of the other recesses **254** of the inner member **253**. While the outer member **255** is described as having a plurality of recesses **254** and the inner member **253** as having a complementary protrusion **256**, it is further contemplated that the inner member **253** may comprise the protrusion **256** and the outer member **255** may comprise the complementary plurality of apertures or recesses **254** for adjusting the sliding mechanism **252**. Furthermore, while a plurality of apertures or recesses **254** and complementary protrusion **256** is described and illustrated as means of adjusting the sliding mechanism **252**, other configurations of the sliding mechanism **252** are contemplated. For example, the sliding mechanism **252** may be configured as a rails system with a friction fit for adjusting the length of the first and second posts **226**, **227**. Alternatively, the sliding mechanism **252** may comprise a mechanical lever and stops for adjusting the length of the posts **226**, **227**. This list of exemplary configurations of the sliding mechanism **252** is not intended to be exhaustive.

(83) The face frame **224** may further comprise a joint **260** defining a pivot point. The joint may also be referred to as a hinge, pivot, swivel, elbow, juncture, bracket or similar fixture capable of pivoting or rotating. The joint **260** may be positioned at any point along the length of the posts **226**, **227** between the upper beam **229** and the lower beam **228**. The joint **260** may define the upper portions **226A**, **227A**, and the lower portions **226B**, **227B** of the posts **226**, **227**, wherein the upper portions **226A**, **227A** are located above the joint **260** (closer to the upper beam **229**) and the lower portions **226B**, **227B** are located below the joint **260** (closer to the lower beam **228**). It is contemplated that the joint **260** may be configured and/or utilized to removably couple the upper portions **226A**, **227A** of the posts **226**, **227** to the lower portions **226B**, **227B** of the posts **226**, **227**.

It is also contemplated that the joint **260** may be configured and/or utilized to removably couple the posts **226**, **227** to the upper beam **229**.

(84) The joint **260** may be configured to allow the lower portions **226B**, **227B** of the posts **226**, **227** to be rotated relative to the upper portions **226A**, **227A** about the pivot point. The joint **260** may serve to manipulate the lower portions **226B**, **227B** of the posts **226**, **227** and/or the lower beam **228** relative to the upper portions **226A**, **227A** of the posts **226**, **227** and/or the upper beam **229**. For example, the joint **260** may be utilized pivot the lower beam **228** distally away from the neck and/or chest of the individual wearing the surgical helmet **220**. This may serve to define a larger volume under/within the surgical garment **12**, **112** to produce better air flow under and/or within the surgical garment **12**, **112**. This may serve to space the lower beam **228** further away from the wearer's chest and prevent the face frame **224** from becoming ensnared on the wearer's chest.

Alternatively, the joint **260** may be utilized pivot the lower beam **228** proximally toward the neck and/or chest of the individual wearing the surgical helmet **220**. This may serve to bring the transparent shield **18**, **118** of the surgical garment **12**, **112** closer the wearer's face improving the wearer's field of view and reducing glare produced on the surface of the transparent shield **18**, **118**.

(85) While FIG. **3** illustrates an exemplary configuration of the surgical helmet **220** including both a sliding mechanism **252** and a joint **260**, it is contemplated that alternative configurations of the surgical helmet **220** may comprise only a sliding mechanism **252** or only a joint **260**.

(86) Referring to FIGS. **4A** and **4B**, a fourth configuration of a surgical helmet **320** is illustrated. Either of the surgical garments **12**, **112** described above may be utilized with the surgical helmet **320** illustrated in FIG. **4A**. Furthermore, it should be appreciated that the various configurations of the surgical helmet **320** may include similar elements that may be identified by reference numerals that are incremented by 100. It should be understood that those elements including reference numerals which are incremented by 100 can have the same features as described above and/or below with regard to other configurations of the surgical helmet **320**.

(87) Similar to as described above, the surgical helmet **320** may comprise a face frame **324** that is coupled to the front nozzle **340** or block **342** of the surgical helmet **320**. The face frame **324** may comprise an upper beam **329** with opposing lateral portions **329A**, **329B**. The upper beam **329** may be positioned to generally span across the forehead of the wearer of the surgical helmet **320**. The upper beam **329** may also comprise a protrusion **346**. The protrusion **346** may be configured to extend distally outward and upward from the upper beam **329**. The protrusion **346** may be positioned proximate the center of the upper beam **329**. While the exemplary configuration of the upper beam **329** illustrated in FIG. **4A** only includes a single protrusion **346**, it is further contemplated that the upper beam **329** may comprise multiple protrusions **346** spaced along the length of the upper beam **329**.

(88) Referring to FIG. **4B**, the face frame **324** may comprise a control housing **350** disposed within the upper beam **329**. The control housing **350** may define a chamber for housing a coupling assembly **362** of the face frame **324**. The coupling assembly **362** may be configured to removably couple the face frame **324** to the front nozzle **340** or block **342** of the surgical the helmet **320**. The coupling assembly may comprise a control member **364** positioned on the exterior of the control housing **350** and manipulatable by the wearer of the surgical helmet assembly **320**. The control member **364** may comprise a button, switch, rotatable dial or other similar control mechanism. The coupling assembly **362** may also comprise a linkage **366** coupling the control member **364** to an engaging member **368**. The engaging member **368** may be configured to move between an engaged position and a disengaged position based on the position of the linkage **366** such that the engaging member **368** selectively engages a corresponding feature **370** on the front nozzle **340** or block **342** of the surgical helmet **320**. For example, the engaging member **368** may comprise a wedge or tooth shape that extends from the control housing **350** toward a surface of the front nozzle **340** or block **342** of the surgical helmet **320**. The surface of the front nozzle **340** or block **342** may comprise a corresponding feature **370**, such as a plurality of teeth **370** configured to engage the engaging

member **368**. When the engaging member **368** is in the engaged position, the engaging member **368** may engage the teeth **370** to lock the face frame **324** in place relative to the front nozzle **340** or block **342** of the surgical helmet **320**. By contrast, when the control member **364** is manipulated by the wearer to move the engaging member **368** to the disengaged position, the engaging member **368** may be disengaged from the teeth **370**, allowing the face frame **324** to slidably move relative to the front nozzle **340** or block **342** of the surgical helmet **320**. This may include sliding the face frame **324** relative to the front nozzle **340** or block **342** to remove and/or couple the face frame **324** to the surgical helmet **320**. The coupling assembly **362** may further comprise a biasing member **365** coupled to the control member **364** and/or the linkage **366** configured to maintain the engaging member **368** in the engaged position absent an external force being applied to the control member **364** to manipulate the linkage **366**.

(89) The coupling assembly **362** may also comprise a coupling arm **374** intended to slidably engage a corresponding channel **372** of the front nozzle **340** or block **342** when the face frame **324** is coupled to the front nozzle **340** or block **342** of the surgical helmet **320**. The channel **372** may define the path on which the face frame **324** may be slid when the control member **364** is manipulated to disengage the engaging member **368** from the teeth **370** of the front nozzle **340** or block **342**.

(90) The face frame **324** may comprise a first post **326** and a second post **327**. Each of the first post **326** and second post **327** may further comprise an upper portion **326A**, **327A**, and a lower portion **326B**, **327B**, respectively. The upper portion **326A**, **327A** of each of the first post **326** and second post **327** may be coupled to the opposing lateral portions **329A**, **329B** of the upper beam **329** of the face frame **324** and extend in a generally downward direction from the upper beam **329**. The first post **326** and second post **327** of the face frame **324** may be constructed from a generally flexible or pliable material.

(91) A lower beam **328** may extend between the opposed lower portions **326B**, **327B** of the posts **326**, **327**. The face frame **324** is formed so that the lower beam **328** is located below and slightly forward of the chin of the person wearing the surgical helmet **320**. The lower beam **328** may be bowed outwardly from the opposed lower portions **326B**, **327B** of posts **326**, **327**. The face frame **324** may extend outwardly from the front nozzle **340** and/or block **342** of the surgical helmet **320** such that the face frame **324** is positioned forward of and generally encircles the face of the wearer when the surgical helmet **320** is secured to the wearer's head.

(92) A plurality of coupling members **348**, such as magnets, hook and loop, metal rivets, snaps, or similar type fasteners, may be mounted to the face frame **324** and configured to align and/or attach one of the transparent shields **18**, **118** of the surgical garments **12**, **112** described above. Each coupling member **348** may be positioned on the face frame **324** proximate to the opposed free ends **326B**, **327B** of the posts **326**, **327** and/or adjacent opposing ends of the lower beam **328**.

Alternatively, the coupling members **348** of the surgical helmet **320** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements **58**, **158** of the transparent shield **18**, **118**, as described above, to releasably secure the surgical garment **12**, **112** to the surgical helmet **320**. For example, the protrusion **346** extending from the upper beam **329** may be replaced with a coupling member **348** configured to cooperate with a complementary attachment element **58**, **158** on the surgical garment **12**, **112**.

(93) Similar to the face frame **224** described above, the face frame **324** of FIG. **4A** may comprise a joint **360**. The joint **360** may be configured to allow the first and second posts **326**, **327** to be rotated relative to the upper beam **329** about the pivot point. The joint **360** may also serve to manipulate the position of the lower beam **328** relative to the upper beam **329**. For example, the joint **360** may be utilized pivot the lower beam **328** distally away from the neck and/or chest of the individual wearing the surgical helmet **320**. This may serve to define a larger volume under/within the surgical garment **12**, **112** to produce better air flow under and/or within the surgical garment **12**, **112**. This may serve to space the lower beam **328** further away from the wearer's chest and prevent

the face frame **324** from becoming ensnared on the wearer's chest. Alternatively, the joint **360** may be utilized to pivot the lower beam **328** proximally toward the neck and/or chest of the individual wearing the surgical helmet **320**. This may serve to bring the transparent shield **18**, **118** of the surgical garment **12**, **112** closer to the wearer's face improving the wearer's field of view and reducing glare produced on the surface of the transparent shield **18**, **118**.

(94) While not illustrated in FIG. **4A**, it is also contemplated that the face frame **324** may comprise a sliding mechanism similar to the sliding mechanism **252** described above.

(95) Referring to FIGS. **5A** to **5D**, a fifth configuration of a surgical helmet **420** including an alignment channel **445** in an upper beam **429** of a face frame **424** is illustrated. This is similar to the surgical apparel system of FIGS. **2A** and **2B** described above. The surgical garment **112** described above may be utilized with the surgical helmet **420** illustrated in FIG. **5A**. Furthermore, it should be appreciated that the various configurations of the surgical helmet **420** may include similar elements that may be identified by reference numerals that are incremented by 100. It should be understood that those elements including reference numerals which are incremented by 100 can have the same features as described above and/or below with regard to other configurations of the surgical helmet.

(96) Referring to FIGS. **5A** and **5B**, the surgical helmet **420** comprises a face frame **424** similar to those described above. The face frame **424** of the surgical helmet **420** may comprise an upper beam **429** with opposing lateral portions **429A**, **429B**. The upper beam **429** may be positioned forward of a shell **432** and coupled to the front nozzle **440** and/or block **442** of the surgical helmet **420** and configured to extend across the front of the surgical helmet **420**. The upper beam **429** may further comprise a control housing **450**. The proximal portion of the control housing **450** may be configured to couple the face frame **424** to the front nozzle **440** and/or block **442** of the surgical helmet **420**. The distal portion of the control housing **450** may include an alignment channel **445**. The alignment channel **445** may be configured to receive a tab **147** disposed on the interior of the surgical garment **112** to align and/or orient the surgical garment **112** relative to the surgical helmet **420**.

(97) The upper beam **429** may further comprise a coupling feature **446** configured to removably engage the transparent shield **118** and/or surgical garment **112**. The coupling feature **446** may comprise a protrusion, magnetic member, ferromagnetic member, hook and loop, or similar coupling mechanism configured to releasably engage the opening **149** in the tab **147** to align and/or couple the surgical garment **112** to the surgical helmet **420**.

(98) The face frame **424** may also comprise a first post **426** and a second post **427** that extend downwardly from the upper beam **429**. Each of the first post **426** and second post **427** may further comprise an upper portion **426A**, **427A**, and a lower portion **426B**, **427B**, respectively. The upper portions **426A**, **427A** of each of the first post **426** and second post **427** may be coupled to the opposing lateral portions **429A**, **429B** of the upper beam **429** of the face frame **424** and extend in a generally downward direction from the upper beam **429**. The first post **426** and second post **427** of the face frame **424** may be constructed from a generally flexible or pliable material. The first and second posts **426**, **427** may be coupled to the upper beam **429**. For example, as illustrated in FIG. **5A**, the first and second posts **426**, **427** may be connected to the opposing ends **429A**, **429B** of the upper beam **429**. The face frame **424** may be constructed from a generally flexible or pliable material.

(99) The face frame **424** may further comprise a lower beam **428** that may extend between the opposed free ends of the posts **426**, **427**. The face frame **424** is formed so that the lower beam **428** is generally located below and slightly forward of the chin of the person wearing the surgical helmet **420**. The lower beam **428** may be bowed outwardly from the free ends of posts **426**, **427**. The face frame **424** may extend outwardly from the upper beam **429** such that the face frame **424** is positioned forward of and generally encircles the face of the wearer when the surgical helmet **420** is secured to the wearer's head.

(100) A plurality of coupling members **448** may be mounted to or within the face frame **424**. The coupling members **448** may comprise a magnetic material and are configured to align and/or attach the transparent shield **118** of the surgical garment **112** to the surgical helmet **420**. Alternatively, the coupling members/complementary attachment elements may be hook and loop fasteners or other suitable implementation. Each coupling member **448** may be positioned on the face frame **424** proximate to the opposed posts **426**, **427** and/or adjacent opposing ends of the lower beam **428**. Alternatively, the coupling members **448** of the surgical helmet **420** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements **158** of the surgical garment **112** to releasably secure the surgical garment **112** to the surgical helmet **420**.

(101) The face frame **424** may further comprise a sliding mechanism **452**. The sliding mechanism **452** may comprise a portion of the first post **426** and second post **427**. As illustrated in FIG. 5A, the sliding mechanism **452** may be positioned along the length of the posts **426**, **427** and may define the upper portions **426A**, **427A** of the posts **426**, **427**, and the lower portions **426B**, **427B** of the posts **426**, **427**. Alternatively, while not illustrated in FIG. 5A, it is contemplated that the sliding mechanism **452** may be positioned proximate where the posts **426**, **427** are coupled to the upper beam **429** and/or may be positioned proximate where the posts **426**, **427** are coupled to the lower beam **428** of the face frame **424**. It is contemplated that the sliding mechanism **452** may be configured and/or utilized to removably couple the upper portions **426A**, **427A** of the posts **426**, **427** to the lower portions **426B**, **427B** of the posts **426**, **427**. It is also contemplated that the sliding mechanism **452** may be configured and/or utilized to removably couple the posts **426**, **427** to the upper beam **429** and/or the lower beam **428**.

(102) As illustrated in FIG. 5A, the protrusion **446** and the coupling member(s) **448** may be positioned on opposing sides of the sliding mechanism **452**, resulting in the distance between the protrusion **446** and the coupling member(s) **448** changing based on the position of the sliding mechanism **452**. This may mean multiple transparent shield **118** and/or surgical garment **112** configurations are required to allow for coupling of the attachment members **158** of the shield **118** to the coupling member(s) **448** of the face frame **424**. However, while not illustrated, it is further contemplated that both the protrusion **446** and the coupling member(s) **448** of the face frame **424** may be on the same side of the sliding mechanism **452**. For example, the coupling member(s) **448** may be positioned above the sliding mechanism **452**. In this configuration, the distance between the protrusion **446** and the coupling member(s) **448** would not change with the position of the sliding mechanism **452**. In this configuration, a single configuration of the transparent shield **118** and/or surgical garment **112** may be coupled to the face frame **424** regardless of the position of the sliding mechanism **452**.

(103) It is also contemplated that the face frame **424** may further comprise a repositionable coupling feature **446**. Referring to FIGS. 5C and 5D, an exemplary configuration of a repositionable coupling feature **446** is illustrated. The upper beam **429** of the face frame **424** may comprise a plurality of apertures **455A**, **455B**, **455C**, **455D** or slots that are spaced vertically on the distal surface of the upper beam **429** from which the coupling feature extends. The coupling feature **446** or protrusion may comprise a corresponding post configured to matingly engage the plurality of apertures **455A**, **455B**, **455C**, **455D** to removably attach the coupling feature **446** to the upper beam **429** of the face frame **424**. For example, the coupling feature **446** may comprise a set or laterally spaced posts that engage one set of the plurality of apertures **455A**, **455B**, **455C**, **455D** at a time to removably attach the coupling feature **446** to the upper beam **429**. The plurality of apertures **455A**, **455B**, **455C**, **455D** in the upper beam **429** may be vertically spaced to correspond to the various positions of the sliding mechanism **452** of the first post **426** and second post **427** to allow the vertical distance to remain constant between the coupling feature **446** of the upper beam **429** and the coupling member(s) **448** of the face frame **424** as the sliding mechanism **452** is manipulated to lengthen or shorten the first post **426** and second post **427**. For example, when the sliding mechanism **452** is fully extended, the coupling feature **446** may be attached to the apertures

455A, as illustrated in FIG. 5C. Alternatively, when the sliding mechanism **452** is fully retracted, the coupling feature **446** may be attached to the apertures **455D**. When the sliding mechanism **452** is positioned at an intermediary position, the coupling feature **446** may be attached to the corresponding apertures **455B**, **455C** to maintain a constant distance between the coupling feature **446** of the upper beam **429** and the coupling member(s) **448** of the face frame **424**. This may allow for a single configuration of the transparent shield **118** and/or surgical garment **112** to be coupled to the face frame **424** irrespective of the position of the sliding mechanism **452** is in.

(104) Referring to FIG. 5B, an exemplary configuration of the sliding mechanism **452** is illustrated. The sliding mechanism **452** may comprise an inner member **453** and an outer member **455** that slidably engage one another to alter and/or manipulate the length of the first post **426** and second post **427**. For example, similar to the sliding mechanism described above, the sliding mechanism **452** may slide between a first position, a second position, or any intermediary position between the first and second positions. When the sliding mechanism **452** is in the first position, the first post **426** and second post **427** may have a first length defining a first distance between the upper beam **429** and the lower beam **428**. This may also be referred to as a first configuration of the face frame **424**. Alternatively, when the sliding mechanism **452** is in the second position, the first post **426** and second post **427** may have a second length defining a second distance between the upper beam **429** and the lower beam **428**. This may also be referred to as a second configuration of the face frame **424**. The face frame **424** may also comprise a plurality of intermediate configurations wherein the sliding mechanism **452** is positioned at a point between the first position and the second position. This allows for the length of the face frame **424** of the surgical helmet **420** to be adjusted based on the specific size and characteristics of the individual wearing the surgical helmet. For example, an individual with a small head and/or a shorter neck may desire a shorter distance between the upper beam **429** and lower beam **428** to prevent the lower beam **428** from contacting and/or becoming ensnared on the wearer's chest. Alternatively, an individual with a larger head and/or a longer neck may desire a longer distance between the upper beam **429** and lower beam **428** to maximize the viewing area provided by the transparent shield **118** of the surgical garment **112** that is coupled to the surgical helmet **420**.

(105) The inner member **453** and the outer member **455** of the sliding mechanism **452** may also comprise complementary features such as a recess **454** and protrusion **456** to define the various lengths of the first post **426** and second post **427**. For example, the inner member **453** may comprise a plurality of recesses **454** spaced along the length of the inner member **453**. The outer member **455** may then comprise at least one protrusion **456** configured to matingly engage the plurality of recesses **454** as the outer member **455** is slid along the length of the inner member **453**. The protrusion **456** when engaging one of the plurality of recesses **454** may be configured to create a friction fit, holding inner member **453** and outer member **455** in place relative to one another until an external force is applied to move the protrusion **456** from the current recess **454** to one of the other recesses **454** of the inner member **453**. The protrusion **456** may be coupled to a biasing member **457** configured to create the friction fit between the protrusion **456** and the recess **454**. The biasing member **457** may bias the protrusion **456** toward the recess **454** to assist in securing the sliding mechanism **452** in place until an external force is applied to the sliding mechanism **452**. While the inner member **453** is described as having a plurality of recesses **454** and the outer member **455** as having a complementary protrusion **456**, it is further contemplated that the inner member **453** may comprise the protrusion **456** and the outer member **455** may comprise the complementary plurality of recesses **454** for adjusting the sliding mechanism **452**. Furthermore, while a plurality of recesses **454** and complementary protrusion **456** is described and illustrated as a means of adjusting the sliding mechanism **452**, other configurations of the sliding mechanism **452** are contemplated. For example, the sliding mechanism **452** may be configured as a rails system with a friction fit for adjusting the length of the posts **426**, **427**. Alternatively, the sliding mechanism **452** may comprise a mechanical lever and stops for adjusting the length of the first and

second posts **426**, **427**. This list of exemplary configurations of the sliding mechanism **452** is not intended to be exhaustive.

(106) While not illustrated in FIG. 5A, it is contemplated that the face frame **424** may comprise a joint similar to those described above in FIG. 3 and/or 4A. It is also contemplated that the surgical helmet assembly **420** of FIG. 5A may comprise a face frame including a coupling assembly similar to the coupling assembly **362** described in FIGS. 4A and 4B.

(107) Referring to FIGS. 6A, 6B, 7A, and 7B, perspective views of various stages of coupling an exemplary configuration of a transparent shield **118**, and by extension the surgical garment **112**, to the face frame **424** of the surgical helmet **420** of FIG. 5A are illustrated. As described above, the transparent shield **118** includes a plurality of configurations, and each configuration may generally comprise a tab **147** defining an opening **149**. While not illustrated, it is also contemplated that the transparent shield **118** may be designed without a tab, and the opening **149** may be defined in a top portion of the transparent shield **118**. Exemplarily configurations of the transparent shield **118A**, **118B** including a tab **147** that is formed as part of the transparent shield **118A**, **118B**, are illustrated in FIGS. 6A, 6B, 7A, and 7B. The upper beam **429** of the surgical helmet **420** may comprise the pair of laterally spaced-apart sidewalls and the proximal surface that defines the alignment channel **445** configured to receive the tab **147** of the transparent shield **118A**, **118B**. The protrusion **446** may extend from the proximal surface of the alignment channel **445** such that the opening **149** of the tab **147** may be disposed over the protrusion **446**. The transparent shield **118** comprises a plurality of attachment elements **158A**, **158B** secured to the periphery of the transparent shield **118A**, **118B**.

(108) To couple the transparent shield **118**, **118A**, **118B**, and by extension the surgical garment **112**, to the surgical helmet **420**, the transparent shield **118**, **118A**, **118B** may be positioned with the opening **149** in the tab **147** above the alignment channel **445** and the protrusion **446** (see FIGS. 6B and 7B). The tab **147**, and by extension the transparent shield **118**, **118A**, **118B**, may then be lowered onto the surgical helmet **420** such that at least a portion of the tab **147** is positioned within the alignment channel **445** between the pair of laterally spaced-apart side walls. The tab **147** should be positioned within the alignment channel **445** such that the protrusion **446** is disposed within the opening **149** of the tab **147**. The lower portion of the transparent shield **118**, **118A**, **118B**, that includes the attachment elements **158**, **158A**, **158B**, may then be manipulated to couple the attachment elements **158**, **158A**, **158B** to the complementary coupling members **448** positioned on the face frame **424**. For example, once the tab **147** is positioned within the alignment channel **445** and the protrusion **446** is disposed within the opening **149** of the tab **147**, the transparent shield **118**, **118A**, **118B** may be pivoted about the protrusion **446** to position the attachment elements **158**, **158A**, **158B** adjacent to the complementary coupling members **448**. This is one example of a method of coupling the surgical garment **112** to the surgical helmet **420**.

(109) As mentioned above, the coupling member **448** comprises one of a ferromagnetic material or a magnetic material and the attachment element **158**, **158A**, **158B** comprises the other of the ferromagnetic material or magnetic material, so that the coupling member **448** and the attachment element **158**, **158A**, **158B** may be magnetically attracted to one another. In the illustrated configurations, the coupling member **448** may comprise a magnetic material, and hence a magnetic field may emanate from or otherwise be generated by the coupling member **448**. When the coupling member **448** is coupled to the attachment element **158**, **158A**, **158B**, the magnetic field surrounding the component comprising the magnetic material will be altered when the component comprising the ferromagnetic material is placed adjacent to it.

(110) As described above, the face frame **424** may comprise a sliding mechanism **452** for varying the length of the posts **426**, **427** of the face frame **424**. As the length of the posts **426**, **427** are manipulated using the sliding mechanism **452**, the distance between the upper beam **429** and the lower beam **428** is altered. In configurations of the face frame **424** where the protrusion **446** of the upper beam **429** and the coupling member(s) **448** of the face frame **424** are on opposing sides of

the sliding mechanism **452**, the distance between the protrusion **446** and the coupling member(s) **448** will also change. This may require that different transparent shield **118**, **118A**, **118B** and/or garment **112** configurations be utilized based on the position of the sliding mechanism **452** and/or the length of the posts **426**, **427**. For example, as illustrated in FIG. **6A**, a first configuration of the transparent shield **118A** is illustrated. The distance between the opening **149** and the attachment element(s) **158A** defines a first distance $X_{sub.1}$ corresponding to a first configuration of the face frame **424A** wherein the sliding mechanism is in a first position (fully retracted). The first distance $X_{sub.1}$ between the opening **149** and the attachment element(s) **158A** corresponds to the distance between the protrusion **446** and the coupling member(s) **448** when the sliding mechanism **452** is in the first position. This will allow the first configuration of the transparent shield **118A**, and by extension the surgical garment **112**, to be coupled to the first configuration of the face frame **424A**, as illustrated in FIG. **6B**.

(111) Alternatively, as illustrated in FIG. **7A**, a second configuration of the transparent shield **118B** is illustrated. The distance between the opening **149** and the attachment element(s) **158B** define a second distance $X_{sub.2}$ corresponding to a second configuration of the face frame **424B** wherein the sliding mechanism **452** is in a second position (fully extended). The second distance $X_{sub.2}$ between the opening **149** and the attachment element(s) **158B** corresponds to the distance between the protrusion **446** and the coupling member(s) **448** when the sliding mechanism **452** is in the second position. This will allow the second configuration of the transparent shield **118B**, and by extension the surgical garment **112**, to be coupled to the second configuration of the face frame **424B**, as illustrated in FIG. **7B**.

(112) While not illustrated, it is further contemplated that additional configurations of the transparent shield **118** including different spacing's between the opening **149** and the attachment element(s) **158** may be required to correspond to additional positions of the sliding mechanism **452** and/or lengths of the posts **426**, **427** of the face frame **424**.

(113) FIGS. **6A-7B** illustrate an exemplary configuration of the face frame **424** including a sliding mechanism **452** for providing single face frame **424** capable of defining a first configuration of the face frame **424A** and a second configuration of the face frame **424B**, wherein the sliding mechanism **452** may be utilized to vary the distance between the protrusion **446** and the coupling member **448**. Each configuration of the face frame **424A**, **424B** may define a distance between the protrusion **446** and the coupling member **448** that correspond to the respective distances between the opening **149** and the attachment element(s) **158** of the various configurations of the transparent shield **118A**, **118B**. While not illustrated in the figures, it is further contemplated that varying the configuration of the face frame may be accomplished in other ways. For example, as described above, it is contemplated that the face frame may be removably coupled to the surgical helmet. In an exemplary configuration of a surgical apparel system wherein the face frame is removably coupled to the surgical helmet, the surgical apparel system may comprise a plurality of removable face frames. Each of the plurality of face frames may define a different distance, such as a coupling distance, between the coupling feature and the coupling member of the particular face frame. The system may further comprise a plurality of transparent shields, wherein each one of the plurality of transparent shields defines a different distance between the opening and the attachment element(s) of the transparent shield corresponding to the one of the distances between the coupling feature and the coupling member of one of the plurality of face frames.

(114) For example, an exemplary configuration of the system may comprise a surgical helmet a first face frame, a second face frame, a first surgical garment, and a second surgical garment. The first face frame may comprise a first upper beam having a first coupling feature, a first pair of posts, and a first lower beam. The first face frame may also comprise a first attachment element at least partially disposed within the first face frame, the first coupling feature and the first attachment element defining a first coupling dimension, such as a height or length. The second face frame may comprise a second upper beam having a second coupling feature, a second pair of posts, and a

second lower beam. The second face frame may also comprise a second attachment element at least partially disposed within the second face frame, the second coupling feature and the second attachment element defining a second coupling dimension. The first surgical garment may comprise a first shield having a first aperture and a first coupling element, wherein the first aperture and the first coupling element define a third coupling dimension. The second surgical garment may comprise a second shield having a second aperture and a second coupling element, wherein the second aperture and the second coupling element define a fourth coupling dimension. The system may be configured such that the first coupling dimension of the first face frame corresponds to the third coupling dimension of said first surgical garment to allow the first surgical garment to couple to the first face frame. The system may further be configured such that the second coupling dimension of the second face frame corresponds to the fourth coupling dimension of the second surgical garment to allow the second surgical garment to couple to the second face frame. This allows for the system to incorporate various face frames with a single surgical helmet, wherein the face frame may be sized to best accommodate the needs of the individual wearing the surgical helmet. For example, as described above, an individual with a small head and/or a shorter neck may prefer a face frame with a smaller coupling dimension, i.e. the shorter distance between the coupling feature and the attachment element of the face frame. A smaller coupling dimension may prevent the face frame from contacting and or snaring on the chest of the individual when wearing the system. This individual may attach a face frame comprising a smaller coupling dimension to the surgical helmet and using the corresponding surgical garment and/or transparent shield.

Alternatively, wherein an individual may prefer a larger coupling dimension, which may provide a larger transparent shield and greater/larger field of view, the individual may attach a face frame comprising a larger coupling dimension to the surgical helmet and use the corresponding surgical garment and/or transparent shield. It is contemplated that the system may comprise any number of face frames, each of which may vary in size, shape, and/or orientation. The system may similarly comprise any number of surgical garments, including a transparent shield, that may also vary by size, shape, and configuration to fit any or all of the various configurations of the face frame.

(115) The system may be configured such that a face frame coupled to the surgical helmet may be swapped out or exchanged for a different configuration of the face frame at any time. For example, the first face frame may be coupled to the helmet for a first procedure to be completed by medical professional number **1**. After completion of the first procedure, the first face frame may be removed from the surgical helmet, and the second face frame may be coupled to the surgical helmet for a second procedure to be completed by medical professional number **2**. The first face frame and second face frame may be swapped out from the surgical helmet to accommodate differing size and/or anatomical features of medical professional number **1** compared to medical professional number **2**. The face frame may also be swapped in the middle of the procedure.

(116) It is also contemplated that the face frames may be swapped out based on type of procedure to be performed. For example, it is contemplated that the first face frame may correspond to a surgical garment that provides better peripheral view based on the size and/or shape of the corresponding surgical garment and transparent shield. However, it is also contemplated that the second face frame may comprise a size and/or shape to accommodate a light being coupled to the surgical helmet or face frame. For example, the second face frame may be configured to provide an attachment feature for coupling a light or other peripheral device to the second face frame. The face frames may also be swapped out and/or removed as part of the sterilization/cleaning process following execution of a medical procedure.

(117) Referring to FIG. **8**, an alternative configuration of the transparent shield **218** of a surgical garment **12** or **112** is illustrated. The transparent shield **218** may comprise an opening **249** configured to receive the protrusion **46**, **146**, **246**, **346**, **446** of one of the helmet assembly **20**, **120**, **220**, **320**, **420** described above. The transparent shield **218** may also comprise a tab **247**, wherein the tab **247** at least partially defines the opening **249**. The transparent shield **218** may further

comprise a plurality of attachment members **258A**, **258B**, **258C**. The attachment members **258A**, **258B**, **258C** may be configured to define varying distances X.sub.1, X.sub.2, X.sub.3, between the opening **249** and the respective attachment members **258A**, **258B**, **258C**. For example, a first attachment member **258A** may define a first distance X.sub.1, a second attachment member **258B** may define a second distance X.sub.2, and a third attachment member **258C** may define a third distance X.sub.3. These distances may correspond to the distance between the protrusion **46**, **146**, **246**, **346**, **446** and the attachment element(s) **48**, **148**, **248**, **348**, **448** of the surgical helmet assembly **20**, **120**, **220**, **320**, **420** based on the position of the sliding mechanism **252**, **452** and/or the joint **260**, **360** of the face frame(s) **24**, **124**, **224**, **324**, **424** described above. This will allow a single configuration of the transparent shield **218**, and by extension the surgical garment, to be coupled to a surgical helmet assembly **20**, **120**, **220**, **320**, **420** including a reconfigurable face frame **24**, **124**, **224**, **324**, **424** described above.

Alternative Features of the Surgical Helmet

(118) Various features and configurations of a surgical helmet **20**, **120** to be used as part of a surgical apparel system **10**, **110** are described above. The surgical helmet **20**, **120** may generally comprise a face frame **24**, **124** that may extend downwardly from the front portion of the surgical helmet **20**, **120** and be positioned generally in front of the wearer's face when the surgical helmet **20**, **120** is worn by a medical professional. As discussed above, the face frame **124** may be reconfigurable, such that the face frame **124** may include pivotable and/or extendable portions. It is also contemplated that the face frame **124** may be removably coupled to the surgical helmet **20**, **120**.

(119) Referring to FIGS. **9A** and **9B**, an alternative configuration of a face frame **524** is illustrated. The face frame **524** may comprise an upper beam **529** with opposing lateral portions **529A**, **529B**. While described as part of the face frame **524**, it is also contemplated that the upper beam **529** may be considered part of the surgical helmet **120**. The upper beam may be positioned forward of the shell **132** of the surgical helmet **120** and configured to extend across the front of the surgical helmet **120**. The upper beam **529** may further and optionally comprise an alignment channel **545** that is defined by a pair of laterally spaced-apart sidewalls and a proximal surface that is positioned proximally from the distal surface of the upper beam **529**. The alignment channel **545** is configured to receive a tab disposed on the interior of the surgical garment to align and/or orient the surgical garment relative to the surgical helmet **120**, as is described in more detail above. While the exemplary configuration of the face frame **524** illustrated in FIGS. **9A** and **9B** comprises an alignment channel **545**, it is contemplated that the face frame **546** may be configured without an alignment channel **545**, similar to the face frame **24** illustrated in FIGS. **1A** and **1B**.

(120) The upper beam **529** may further comprise a coupling feature **546**, such as a protrusion, configured to removably engage one of the transparent shields and/or surgical garments described above. The coupling feature **546** may comprise a protrusion, magnetic member, ferromagnetic member, hook and loop, or similar coupling mechanism configured to releasably engage the surgical garment to align and/or couple the surgical garment to the surgical helmet **120**. For example, as illustrated in FIG. **9A**, the coupling feature **546** is realized as a protrusion **546** extending from the alignment channel **545** of the upper beam **529**. The combination of the spaced-apart sidewalls of the alignment channel **545** and the coupling feature **146** may serve to align and/or orient the transparent shield and/or the surgical garment relative to the surgical helmet **120**. More specifically, the spaced-apart sidewalls of the alignment channel **545** may serve to guide the tab of the transparent shield and/or the surgical garment into engagement with the coupling feature **546** as the surgical garment is placed over the surgical helmet **120**.

(121) The upper beam **529** may comprise a mounting member **556**, **558** positioned at opposing ends of the lateral portions **529A**, **529B** of the upper beam **529**. As illustrated in FIGS. **9A** and **9B**, the mounting feature may comprise a first mounting feature **556** and a second mounting feature **558**. For example, the first mounting feature **556** may comprise a post **556** extending from the

distal surface of the upper beam 529, where the post 556 comprises a fastening feature 556A. The second mounting feature 558 may comprise a post 558 extending from the distal surface of the upper beam 529. While the mounting feature(s) 556, 558 illustrated in FIGS. 9A and 9B include a pair of posts 556, 558, it is also contemplated that the mounting feature(s) 556, 558 may be configured as a magnet, a hook, a hook and loop fastener, one or more apertures, a threaded fastener, or the like.

(122) The face frame 524 may comprise a first post 526 and a second post 527. The first and second posts 526, 527 may be coupled to the upper beam 529 by a mounting bracket 553. The mounting bracket 553 may be formed from generally flexible material and be configured to extend from the top end of the first and second posts 526, 527. The mounting bracket 553 may comprise a face frame coupler 552, 554 corresponding to the mounting feature(s) 556, 558 of the upper beam 529 to couple the first and second posts 526, 527 to the upper beam 529. For example, as illustrated in FIGS. 9A and 9B, the mounting bracket 553 may comprise a first aperture 552 and a second aperture 554. The first aperture 552 may be configured to couple with the first mounting feature 556. For example, the first aperture 552 may define an opening in the bracket 553 that is shaped to allow the fastening feature 556A of the first mounting feature 556 to be inserted through the first aperture 552, and then the first post 556 may be seated into a smaller portion of the first aperture 552 such that the fastening feature 556A prevents the bracket 553 from sliding off of the first post 556. The second aperture 554 in the bracket 553 may correspond to the second mounting feature 558, wherein the second aperture 554 and the second mounting feature 558 are arranged and configured to secure the first aperture 552 in place relative to the first mounting feature 556.

(123) Where the upper beam 529 is configured to extend across the front of the surgical helmet 120, the mounting feature(s) 556, 558 and the mounting bracket(s) 553 of the first and second posts 526, 527 may be configured to couple with one another to attach the first and second posts 526, 527 to the corresponding lateral portions 529A, 529B of the upper beam 529. For example, as illustrated in FIGS. 9A and 9B, the first and second posts 526, 527 may be connected to opposing ends 529A, 529B of the upper beam 529 by inserting the mounting feature(s) 556, 558 into the corresponding face frame coupler 552, 554 of the mounting bracket 553. For example, to secure the post 526, 527 to the upper beam 529, the mounting bracket 553 may be flexed to manipulate the second aperture 554 away from the second mounting feature 558 to position the first mounting feature 556 within the first aperture 552. Once the first mounting feature 556 is positioned within the first aperture 552, the bracket 553 may be allowed to return to its normal state/orientation, positioning the second mounting feature 558 within the second aperture 554. Alternatively, to remove the post 526, 527 from the upper beam 529, the mounting bracket 553 may be flexed to disengage the second aperture 554 from the second mounting feature 558, allowing the mounting bracket 553 to be manipulated to disengage the first mounting feature 556 from the first aperture 552.

(124) The face frame 524 may further comprise a lower beam 528 that may extend between the opposed free ends of the first and second posts 526, 527. The face frame 524 is formed so that the lower beam 528 is located below and slightly forward of the chin of the person wearing the surgical helmet 120. The lower beam 528 may be bowed outwardly from the free ends of the posts 526, 527. The face frame 524 may extend outwardly from the upper beam 529 such that the face frame 524 is positioned forward of and generally encircles the face of the wearer when the surgical helmet 120 is secured to the wearer's head. Collectively, the combination of the first and second posts 526, 527 and the lower beam 528 may also be referred to as a chin bar, as they generally define an opening positioned in front of the wearer's face and extend below the wearer's chin when the surgical helmet is positioned on top of the wearer's head.

(125) Similar to the chin bars and/or face frames described above, the face frame 524 may comprise a plurality of coupling members 548 that may be mounted to or disposed within the face frame 524. The coupling members 548 comprise magnetic material and are configured to align

and/or attach one of the various transparent shields and/or the surgical garments described above to the surgical helmet **120**. Each coupling member **548** may be positioned on the face frame **524** proximate to the opposed first and second posts **526**, **527** and/or adjacent opposing ends of the lower beam **528**. Alternatively, the coupling members **548** of the surgical helmet **120** could be arranged or otherwise configured in any suitable way to cooperate with the complementary attachment elements of the surgical garment to releasably secure the surgical garment to the face frame **524** of the surgical helmet **120**. It is contemplated that other types of coupling members **548** may be used in place of and/or in addition to those comprising magnetic materials, such as with hook and loop fasteners, snaps, coupling members comprising ferromagnetic materials, or similar type fasteners. Other configurations are contemplated. Exemplary configurations of coupling members are disclosed in U.S. Pat. No. 10,420,386, which is hereby incorporated by reference in its entirety.

(126) Referring to FIG. **9A**, an exemplary configuration of a coupling member **548** positioned within the face frame **524** is illustrated. The coupling member **548** may comprise a distal surface. The face frame **524** may comprise a recess configured to receive the coupling feature **548**. For example, the coupling member **548** may be positioned within the recess of the face frame **524**, such that the distal surface of the coupling member **548** is positioned proximally to a distal surface of the face frame **524**.

(127) While the exemplary face frame **524** illustrated in FIGS. **9A** and **9B** includes many of the features illustrated in and described with respect to FIGS. **2A** and **2B**, it is also contemplated that the face frame **524** may be configured to include features similar to those illustrated in and described with respect to FIGS. **1A** and **1B**. The mounting feature(s) **556**, **558** and the mounting bracket **553** are not intended to be limited to a specific configuration, but are intended be an exemplary means of removably coupling any variation or configuration of a chin bar and/or face frame to **524** to a surgical helmet **120**.

(128) Referring to FIGS. **10A** and **10B**, an alternative configuration of the face frame **624** including a mounting feature **656** and mounting bracket **653** for removably coupling the face frame **624** to a surgical helmet **120** is illustrated. A number of components of the face frame **624** illustrated in FIGS. **10A** and **10B** comprise the same last two digits as various components described above with regard to alternative features and/or configurations of the surgical helmet, face frame, and/or surgical garment. It should be understood that the components of the face frame **624** and surgical helmet **120** illustrated in FIGS. **10A** and **10B** having the same last two digits as other components described above may comprise any and/or all of the same features or characteristics of the similar components described above.

(129) FIGS. **10A** and **10B** illustrate an alternative means for removably coupling a face frame **624** to a surgical helmet **120**. The face frame **624** may comprise an upper beam **629** with opposing lateral portions **629A**, **629B**. The upper beam may be positioned forward of the shell **132** of the surgical helmet **120** and configured to extend across the front of the surgical helmet **120**.

(130) The upper beam **629** may comprise a mounting feature **656** positioned at opposing ends of the lateral portions **629A**, **629B** of the upper beam **629**. As illustrated in FIGS. **10A** and **10B**, the mounting feature may comprise a mounting feature **656**. For example, the mounting feature **656** may comprise a magnet **656** disposed on the distal surface of the upper beam **629**. While the mounting feature **656** is described as being a magnet, it is also contemplated that the mounting feature **656** may be configured as a ferromagnetic material, a hook and loop fastener, or the like.

(131) The face frame **624** may comprise a first post **626** and a second post **627**. The first and second posts **626**, **627** may be coupled to the upper beam **629** by a mounting bracket **653**. The mounting bracket **653** may comprise a face frame coupler **652** corresponding to the mounting feature **656** of the upper beam **629** to couple the first and second posts **626**, **627** to the upper beam **629**. For example, as illustrated in FIGS. **10A** and **10B**, the mounting bracket **653** may comprise a face frame coupler **652** comprising a ferromagnetic material. Alternatively, it is also contemplated

that the face frame coupler **652** may comprise a magnet. The face frame coupler **652** may be configured to couple with the mounting feature **656** of the upper beam **629**. For example, the face frame coupler **652** may be one of a ferromagnetic material or a magnet disposed on or within the bracket **653** and the mounting feature **656** may comprise the other of a ferromagnetic material or a magnet, such that a magnetic attraction may be formed between the face frame coupler **652** and the mounting feature **656**, allowing for coupling of the posts **626**, **627** to the upper beam **629**. The posts **626**, **627** may be formed of a flexible material to allow the posts **626**, **627** to be manipulated to position the face frame coupler **652** adjacent the mounting feature **656** of the upper beam **629**.

(132) Referring to FIGS. **11A** and **11B**, an alternative configuration of a face frame **724** including a mounting feature **756** and mounting bracket **753** for removably coupling the face frame **724** to a surgical helmet **120** is illustrated. A number of components of the face frame **724** illustrated in FIGS. **11A** and **11B** comprise the same last two digits as various components described above with regard to alternative features and/or configurations of the surgical helmet, face frame, and/or surgical garment. It should be understood that the components of the face frame **724** and surgical helmet **120** illustrated in FIGS. **11A** and **11B** having the same last two digits as other components described above may comprise any and/or all of the same features or characteristics of the similar components described above.

(133) FIGS. **11A** and **11B** illustrate an alternative means for removably coupling a face frame **724** to a surgical helmet **120**. The face frame **724** may comprise an upper beam **729** with opposing lateral portions **729A**, **729B**. The upper beam may be positioned forward of the shell **132** of the surgical helmet **120** and configured to extend across the front of the surgical helmet **120**.

(134) The upper beam **729** may comprise a mounting feature **756** positioned at opposing ends of the lateral portions **729A**, **729B** of the upper beam **729**. As illustrated in FIGS. **11A** and **11B**, the mounting feature **756** may comprise an aperture. For example, the mounting feature **756** may comprise an aperture **756** defined by the distal surface of the upper beam **729**. While the mounting feature **756** is described as being an aperture, it is also contemplated that the mounting feature **756** may be configured as a loop, slot, or the like.

(135) The face frame **724** may comprise a first post **726** and a second post **727**. The first and second posts **726**, **727** may be coupled to the upper beam **729** by a mounting bracket **753**. The mounting bracket **753** may comprise a face frame coupler **752** corresponding to the mounting feature **756** of the upper beam **729** and configured to couple the first and second posts **726**, **727** to the upper beam **729**. For example, as illustrated in FIGS. **11A** and **11B**, the mounting bracket **753** may comprise a face frame coupler **752** comprising a post. Alternatively, it is also contemplated that the face frame coupler **752** may comprise a hook, clasp, snap, pin, or other similar fastener. The face frame coupler **752** may be configured to couple with the mounting feature **756** of the upper beam **729**. For example, as illustrated in FIG. **11B**, the face frame coupler **752** may be a post extending from the bracket **753**, and the mounting feature **756** may comprise an aperture configured to receive the face frame coupler **752**. The face frame coupler **752** and the mounting feature **756** may be sized to create a friction fit to allow for coupling of the posts **726**, **727** to the upper beam **729**. The posts **726**, **727** may be formed of a flexible material to allow the posts **726**, **727** to be manipulated to position the face frame coupler **752** within the mounting feature **756** of the upper beam **729**. It is also contemplated that the face frame coupler **752** and the mounting feature **756** may comprise complementary magnetic and ferromagnetic materials to create a magnetic attraction between the face frame coupler **752** and the mounting feature **756**. For example, the mounting feature **756** may comprise an aperture defined by a magnets disposed in the upper beam **729**, and the face frame coupler **752** may comprise a post made of a magnetic material such that the magnetic attraction between the face frame coupler **752** and the mounting feature **756** holds the post within the aperture when the posts **726**, **727** are coupled to the upper beam **729**.

(136) Referring to FIGS. **12A-12E**, an alternative configuration of a face frame **824** for use with a surgical helmet is illustrated. A number of components of the face frame **824** illustrated in FIGS.

12A-12E comprise the same last two digits as various components described above with regard to alternative features and/or configurations of the surgical helmet, face frame, and/or surgical garment. It should be understood that the components of the face frame **824** and surgical helmet **120** illustrated in FIGS. **12A-12E** having the same last two digits as other components described above may comprise any and/or all of the same features or characteristics of the similar components described above.

(137) The face frame **824** may be configured to be removably coupled to the surgical helmet **120**. The surgical helmet **120** may comprise one more coupling features **856**, **858** disposed on and/or within the shell **132** of the surgical helmet **120**. More specifically, a first coupling feature **856** may be disposed on and/or within front bellows **136** or front nozzle **140** of the shell **132** of the surgical helmet **120**. The surgical helmet **120** may also comprise a second coupling feature **858** that may be disposed on the shell **132**. More specifically, the second coupling feature **858** may be disposed on a support member extending from the shell **132** of the surgical helmet **120**. The coupling features **856**, **858** may comprise a magnet. Alternatively, the coupling features **856**, **858** may comprise a tab, slot, channel, hook and loop, post, aperture, or similar fastening means.

(138) As illustrated in FIGS. **12A-12E**, the face frame **824** may comprise an upper beam **829** with opposing lateral portions **829A**, **829B**. The upper beam **829** may be positioned forward of the shell **132** of the surgical helmet **120** and configured to extend across the front of the surgical helmet **120**. The upper beam **829** may comprise one more fastening members **852** disposed on or proximate the proximal surface of the upper beam **829**. The fastening members **852** may comprise a disk, rivet, post, tab, hook and loop, or similar fastening means that may be configured to be removably coupled with the complementary coupling feature **856** of the surgical helmet **120**. For example, the fastening members **852** may comprise one of a ferromagnetic rivet and a magnet, and the coupling feature **856** may comprise the other of a magnet and a ferromagnetic rivet configured to create a magnetic attraction to removably couple the face frame **824** to the surgical helmet **120**.

Alternatively, the coupling feature **856** of the surgical helmet **120** may comprise an aperture or slot and the fastening member **852** of the face frame **824** may comprise a complementary-sized post or tab configured to be inserted within the coupling feature to removably couple the face frame **824** to the surgical helmet **120**. It is further contemplated that a combination of the exemplary coupling features **856** and exemplary fastening members **852** may be utilized. For example, the coupling feature **856** of the surgical helmet **120** may comprise a magnet disposed within a recess, and the fastening members **852** may comprise a post constructed from a ferromagnetic material configured to fit within the recess when the fastening members **852** are positioned adjacent the coupling feature **856** to removably couple the face frame **824** to the surgical helmet **120**.

(139) The face frame **824** may comprise a first post **826** and a second post **827**. The first and second posts **826**, **827** may extend from the upper beam **829**. The face frame **824** may further comprise a lower beam **828** that may extend between the opposed free ends of the first and second posts **826**, **827** opposite the upper beam **829**.

(140) The face frame **824** may further comprise a bracket **853** extending from each of the opposing lateral portions **829A**, **829B** of the upper beam **829**. The bracket **853** may be configured to extend in a generally proximal direction from the upper beam **829** toward the rear of the surgical helmet **120**.

(141) The face frame **824** may further comprise a mass **857** coupled to the proximal portion of the bracket **853** opposite the distal portion that is coupled to the upper beam **829**. The mass **857** may also be referred to as a counterweight, counterbalance, ballast, weight, or stabilizer. The mass **857** may comprise a dens material, such as brass, iron, copper, tungsten or steel. It is contemplated that the mass **857** may be permanently or removably coupled to the bracket **853**. For example, the mass **857** may be permanently coupled to the bracket **853** using an epoxy or adhesive. It is also contemplated that the mass **857** may be removably coupled to the bracket **853** using a screw, bolt, nut, friction fit or another similar fastening technique or device. The mass **857** may be weighted

and positioned relative to the face frame **824** to offset the weight of upper beam **829**, posts **826**, **827**, and the lower beam **828** of the face frame **824** when the face frame **824** is coupled to the surgical helmet **120**. The purpose of the mass **857** is to balance the center of gravity of the surgical helmet **120**. For example, it is contemplated that the surgical helmet **120** may be designed and configured to be used by a medical professional without a face frame **824**. Therefore, it is contemplated that the surgical helmet **120** may be designed and configured to position the center of gravity of the surgical helmet **120** proximate the center of the medical professional's head without the face frame **824**. Typically, when coupling additional components to the surgical helmet, such as the face frame **824**, the center of gravity is likely to be shifted laterally in the direction where the component is added. For example, when attaching the face frame **824** to the front of the surgical helmet **120**, the center of gravity of the surgical helmet **120** would be shifted forward, moving the center of gravity away from the center of the wearer's head. The center of gravity not being positioned over the center of the wearer's head is likely to cause additional strain and/or physical exertion by the wearer to support the surgical helmet **120** with the face frame **824**. However, the mass **857** coupled to the proximal end of the bracket **853** may comprise a specific weight relative to the weight of the face frame **824** and or the surgical helmet to offset the shift in the center of gravity of the surgical helmet **120** when the face frame **824** is attached. It is also contemplated that the mass **857** may be positioned at a specific distance along the bracket **853** from the upper beam **829**, such that the combination of the weight and the position of the mass **857** relative to the surgical helmet **120** will balance the center of gravity of the surgical helmet **120** in the lateral directions. While the center of gravity of the surgical helmet **120** may move vertically relative to the wearer's head, the center of gravity will remain generally centered over the wearer's head. While the center of gravity of the surgical helmet **120** is generally described relative to the center of the wearer's head, it is contemplated that the preferred location for the center of gravity of the surgical helmet **120** may not be directly over the center of the wearer's head based on the preferred and most beneficial location for loading to be placed on the wearer's head, neck, and/or spine. Describing the use of a mass **857** to maintain the center of gravity over the center of the wearer's head is only for the purpose of explanation and it is contemplated that the mass **857** may be utilized to maintain the center of gravity at any preferred or optimal location over the wearer's head, neck, and or spine.

(142) Referring to FIGS. **12B** and **12C**, the mass **857** may further comprise a fastening member **854**. The fastening member **854** may comprise a disk, rivet, post, tab, hook and loop, or similar fastening means that may be configured to be removably coupled with a complementary second coupling feature **858** disposed on or within the shell **132** of the surgical helmet **120**. For example, the fastening members **854** may comprise one of a ferromagnetic rivet and a magnet, and the second coupling feature **858** may comprise the other of a magnet and a ferromagnetic rivet configured to create a magnetic attraction to removably couple the bracket **853**, the mass **857**, and/or the face frame **824** to the surgical helmet **120**. It is also contemplated that the second coupling feature **858** may comprise a magnet and the mass **857** may be constructed of a ferromagnetic material such that the mass **857** may couple directly to the second coupling feature **858** without need for a separate fastening member **854**. Alternatively, the coupling feature **858** of the surgical helmet **120** may comprise an aperture or slot, and the fastening member **854** of the mass **857** may comprise a complementary-sized post or tab configured to be inserted within the second coupling feature **858** to removably couple the bracket **853**, the mass **857**, and/or the face frame **824** to the surgical helmet **120**. It is further contemplated that a combination of the exemplary coupling features **858** and exemplary fastening members **854** may be utilized. For example, the second coupling feature **858** of the surgical helmet **120** may comprise a magnet disposed within a recess and the fastening members **854** may comprise a post constructed from a ferromagnetic material configured to fit within the recess when the fastening members **854** are positioned adjacent the coupling feature **858** to removably couple the bracket **853**, the mass **857**,

and/or the face frame **824** to the surgical helmet **120**.

(143) It is also contemplated that the mass **857** may be repositionable relative to the surgical helmet **120** to manipulate the location of the center of gravity of the surgical helmet **120** when the face frame **824** is coupled to the surgical helmet **120**. Referring to FIGS. **12D** and **12E**, exemplary features of the face frame **824** and/or surgical helmet **120** for repositioning the mass **857** are illustrated. For example, as illustrated in FIG. **12D**, the bracket **853** may comprise a plurality of apertures **855A**, **855B**, **855C**. As described above, the mass **857** may comprise a fastening member **854**. In addition to coupling the mass **857** to the coupling feature **858** of the surgical helmet **120**, the fastening member **854** may be configured to be removably coupled to the mass **857** to the bracket **853**. For example, the mass may comprise a post that may be inserted through one of the plurality of apertures **855A**, **855B**, **855C** in the bracket **853** and the fastening member **854** may comprise a corresponding opening configured to receive the post of the mass **857**. Alternatively, the mass **857** and fastening member **854** may comprise corresponding snap fit, magnetic, or similar coupling features that allow the mass **857** and fastening member **854** to removably couple to the bracket **853**.

(144) To accommodate the fastening member **854** being repositionable on the bracket **853**, the coupling feature **858** of the surgical helmet **120** may similarly be repositionable. For example, as illustrated in FIG. **12D**, the coupling feature **858** may be slidably mounted to the surgical helmet **120**. For example, the surgical helmet **120** may comprise a slot **859** that the coupling feature **858** may be slid within. The slot **859** may further comprise one or more grooves **860A**, **860B**, **860C** or notches spaced along the slot **859**. The grooves **860A**, **860B**, **860C** may be configured to hold the coupling feature **858** at that position in the slot of a particular groove **860A**, **860B**, **860C** until a force is applied to the coupling feature **858** to move the coupling feature **858** to a different groove **860A**, **860B**, **860C**. For example, as illustrated in FIG. **12D**, the coupling feature **858** is seated in a first groove **860A** of the slot **859**. The position of the grooves **860A**, **860B**, **860C** in the slot **859** may coincide with the various apertures **855A**, **855B**, **855C** in the bracket **853**, such that when the mass **857** and fastening member **854** are positioned in the first aperture **855A**, the coupling feature **858** may be seated in the first groove **860A** of the slot **859** to allow the fastening member **854** to removably couple with the coupling feature **858** when attaching the face frame **824** to the surgical helmet **120**. Alternatively, when the mass **857** and fastening member **854** are positioned in the third aperture **855C**, the coupling feature **858** may be seated in the third groove **860C** of the slot **859** to allow the fastening member **854** to removably couple with the coupling feature **858** when attaching the face frame **824** to the surgical helmet **120**. While only three grooves **860A**, **860B**, **860C** and three apertures **855A**, **855B**, **855C** are illustrated in FIG. **12D**, it is the surgical helmet **120** and/or the face frame **824** may be configured with any number of grooves **860** and corresponding apertures **855** based on the desired placement of the fastening member **854** and the coupling feature **858**.

(145) Referring to FIG. **12E**, an alternative configuration for a repositionable mass **857** is illustrated. As opposed to the surgical helmet **120** including a slot or track for manipulating the position of the coupling feature **858** as illustrated in FIG. **12D**, the configuration of the surgical helmet **120** in FIG. **12D** includes a plurality of coupling features **858A**, **858B**, **858C**. The coupling features **858A**, **858B**, **858C** may be spaced about the surgical helmet **120** to define different attachment points for coupling the face frame **824**.

(146) Similar to the face frame **824** described above, the bracket **853** may comprise a plurality of apertures **855A**, **855B**, **855C** that allow the mass **857** and/or the fastening member **854** to be repositioned along the length of the bracket **853**. The plurality of apertures **855A**, **855B**, **855C** may be spaced along the bracket **853** correspond to the position of the coupling features **858A**, **858B**, **858C** of the surgical helmet **120**, such that the plurality of apertures **855A**, **855B**, **855C** will position the fastening member **854** at a location of the bracket **853** to allow for coupling of the fastening member with one of the coupling features **858A**, **858B**, **858C** of the surgical helmet **120** when coupling the face frame **824** to the surgical helmet **120**.

(147) While not illustrated in the Figures, it is also contemplated that the surgical helmet **120** and the bracket **853** may be configured with a single coupling feature **858** and corresponding fastening member **854**, as is illustrated in FIGS. **12A** and **12B**. The bracket **853** may then comprise a plurality of coupling member, such as a magnet, ferromagnetic coupler, hook and loop fastener, or similar fastener spaced along the length of the bracket **853** at different distances and/or positions. The mass **857** may then comprise a corresponding coupling member to for removably coupling the mass **857** to the any of the plurality of coupling members on the bracket **853**. This would allow the mass **857** to be repositioned along the length of the bracket **853** to manipulate the location of the center of gravity of the surgical helmet **120** when the face frame **824** is coupled to the surgical helmet **120**.

(148) Referring to FIGS. **13A-13D**, an exemplary configuration of a face shield **318A** of a surgical garment is illustrated. The face shield **318A** may be configured to be used with any of the surgical helmets **20**, **120**, and/or as a component of the various surgical garments **12**, **112** described above. In configurations of the surgical helmet **20**, **120** including a face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**, the face shield **18**, **118**, **218** may be at least partially coupled to the surgical helmet **20**, **120** via the coupling members **48**, **148**, **248**, **348**, **448**, **548**, **648**, **748**, **848** disposed on and/or within a portion of the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**. In those configurations of the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824** where the coupling members **48**, **148**, **248**, **348**, **448**, **548**, **648**, **748**, **848** disposed on and/or within the post **26/27**, **126/127**, **226/227**, **326/327**, **426/427**, **526/527**, **626/627**, **726/727**, **826/827**, the coupling members **48**, **148**, **248**, **348**, **448**, **548**, **648**, **748**, **848** also serve to form the curve of the lower portion of the face shield **18**, **118**, **218**. However, as described above, it is also contemplated that there are configurations of the surgical helmet **20**, **120** where all or a portion of the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824** may be removably coupled to the surgical helmet **20**, **120**. In these configurations, the coupling members **48**, **148**, **248**, **348**, **448**, **548**, **648**, **748**, **848** disposed on and/or within a portion of the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824** may not be present and/or available for coupling the face shield **18**, **118**, **218** to the surgical helmet **20**, **120**. Therefore, an alternative face shield **318A** and/or surgical garment configuration may be required for use with the surgical helmet **20**, **120** that does not require the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**.

(149) FIGS. **13A-13D** illustrate an exemplary configuration of the face shield **318A** of a surgical garment **12**, **112** that may couple to a surgical helmet **20**, **120** that does not include a face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**. The face shield may also be referred to as a transparent shield **318A**. The face shield **318A** may comprise a shield adjustment mechanism **380A**, **380B** configured to manipulate the curvature of the face shield **318A**. The shield adjustment mechanism **380A**, **380B** may be coupled to the lower portion of the face shield **318A**. The shield adjustment mechanism **380A**, **380B** may be configured to extend and contract to manipulate the curvature of the lower portion of the face shield **318A**. For example, as illustrated in FIG. **13A**, the face shield **318A** may comprise one or more shield adjustment mechanisms **380A**, **380B** positioned on the lower portion of the face shield **318A**.

(150) The shield adjustment mechanism **380A**, **380B** may be configured as a mechanical actuator similar to a hydraulic cylinder. The shield adjustment mechanism **380A**, **380B** may comprise a cylinder **384A**, **384B** and a piston or rod portion **382A**, **382B** at least partially disposed within the cylinder **384A**, **384B**. Opposing ends of the shield adjustment mechanism **380A**, **380B** may be coupled to an interior surface (the surface of the face shield **318A** on the wearer side of the surgical garment **12**, **112**) of the face shield **318** by a hinge **386**, a pivot, a joint, or similar pivoting connection. As the shield adjustment mechanism **380A**, **380B** expands and contracts, the curvature of the face shield **318A** is adjusted.

(151) In operation, the wearer may manipulate the curvature of the face shield **318A** by flexing the face shield **318A** and/or manipulating the shield adjustment mechanism **380A**, **380B**. For example, the shield adjustment mechanism **380A**, **380B** may be configured such that the piston **382A**, **382B** may resist movement within the cylinder **384A**, **384B** absent the application of an outside force. In

this scenario, with the face shield **318A** starting from a first state, such as a flat state, as illustrated in FIG. **13B**, the user may flex the lower portion of the face shield **318A** to adjust the curvature of the face shield **318A**. As the wearer flexes the face shield **318A**, the piston **382A**, **382B** will retract within the cylinder **384A**, **384B**, shortening the distance between the opposing ends of the shield adjustment mechanism **380A**, **380B**. The shield adjustment mechanism **380A**, **380B** may be configured such that, once the wearer discontinues applying a force to the face shield **318A** and/or the shield adjustment mechanism **380A**, **380B**, the cylinder **384A**, **384B** will hold the piston **382A**, **382B** in the final position unless additional force is applied to further adjust the curvature of the shield. FIG. **13C** illustrates an exemplary curvature of the face shield **318A** created by manipulating the face shield **318A** and/or the shield adjustment mechanism **380A**, **380B**. FIG. **13D** illustrates an exemplary configuration of a face shield **318A** including a pair of shield adjustment mechanisms **380A**, **380B** for manipulating the curvature of the face shield **318A**.

(152) While the shield adjustment mechanism **380A**, **380B** is illustrated as a piston and cylinder, other configurations and/or actuating devices are contemplated. For example, the shield adjustment mechanism **380A**, **380B** may be configured as a ratchet and strap mechanism similar to a cable tie. In this exemplary configuration, the strap may comprise an integrated gear rack and the strap may be inserted within the ratchet such that the gear rack engages the lever of the ratchet. The wearer may pull the strap through the ratchet, shortening the length of the strap and defining a curvature in the face shield **318A**. The ratchet and strap mechanism may be configured such that, once the strap is pulled through the ratchet, the strap cannot be released. Alternatively, the ratchet may comprise a release button or lever configured to disengage the lever of the ratchet from the gear rack of the strap. These are only but a few examples of potential configurations of the shield adjustment mechanism **380A**, **380B** and are not intended to be limiting.

(153) The face shield **318A** may also comprise one or more attachment elements **358**. Exemplary configurations of attachment elements are disclosed in U.S. Pat. No. 10,420,386, which is hereby incorporated by reference in its entirety. As illustrated in FIGS. **13A-13D**, the attachment element(s) **358** may be disposed on the upper portion of the face shield **318A** in order to couple to a surgical helmet **20**, **120** with or without a face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**. For example, the surgical helmet **20**, **120** and/or the face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824** may be modified to position the corresponding coupling members **48**, **148**, **248**, **348**, **448**, **548**, **648**, **748**, **848** of the surgical helmet **20**, **120** on the upper beam **29**, **129**, **229**, **329**, **429**, **529**, **629**, **729**, **829** or similar location on the surgical helmet **20**, **120** so that the face shield **318A** can be coupled to the surgical helmet **20**, **120** with or without a face frame **24**, **124**, **324**, **424**, **524**, **624**, **724**, **824**.

(154) The face shield **318A** may also comprise a tab **347** configured to define an opening **349**. While the face shield **318A** illustrated in FIGS. **13A-13D** includes a tab **347**, it is also contemplated that the face shield **318A** may be configured without a tab **349**. Where the face shield **318A** does not include a tab, the opening **349** may be defined in the face shield **318A**, similar to the face shield **18** of the surgical garment **12** illustrated in FIG. **1A**. Alternatively, it is also contemplated that the face shield **318A** may be configured without an opening **349**. For example, a tab separate from the face shield **318A** may be disposed on the interior of the surgical garment **12**, **112** and configured to couple with the surgical helmet **20**, **120**. Alternatively, neither the face shield **318A** nor the surgical garment **12**, **112** may comprise an opening, and the face shield **318A** and/or surgical garment **12**, **112** may be coupled to the surgical helmet exclusively by the attachment elements **58**, **158**, **258**, **358**.

Clauses to Alternative Configurations

(155) I. A surgical helmet assembly for use with a surgical garment including a coupling element, said assembly comprising: a surgical helmet comprising: a headband for securing said surgical helmet to a wearer; a ventilation unit supported by the headband; a face frame coupled to said surgical helmet, said face frame comprising: an upper beam comprising opposing regions; a first post and a second post, each having a first end and a second end, said first end of each of said first

and said second posts being coupled to said opposing regions of said upper beam; a lower beam connecting said second ends of said first and said second posts; and wherein each of said first post and said second post comprises a top member and a bottom member, said bottom member being movably coupled to said top member to provide a variable length of each of said first post and said second post to vary the distance between said upper beam and said lower beam.

(156) II. The surgical helmet assembly of clause I, wherein said top member and said bottom member of each of said first post and said second post are connected by a sliding mechanism.

(157) III. The surgical helmet assembly of clause II, wherein said sliding mechanism further comprises: a plurality of recesses spaced along a length of an inner member of said sliding mechanism; a detent disposed on an outer member of said sliding mechanism and configured to removably engage each of said plurality of recesses as said outer member is slid along said length of said inner member.

(158) IV. The surgical helmet assembly of clause III, wherein said outer member of said sliding mechanism further comprises a biasing member coupled to said detent for creating a friction fit between said detent and one of said plurality of recesses of which the detent is engaging.

(159) V. The surgical helmet assembly of clause I, further comprising a coupling member disposed on each of said opposing regions of said upper beam for coupling said first end of each of said first and said second posts to said upper beam;

(160) VI. The surgical helmet assembly of clause V, wherein said coupling member defines a first pivot with a coupling feature of said first and said second posts that is configured to allow each of said first and said second posts to rotate relative to said upper beam.

(161) VII. The surgical helmet assembly of clause I, wherein said top member of each of said first and second posts comprises a first joint defining a second pivot to allow said lower beam to rotate about said second pivot defined by said first joint.

(162) VIII. The surgical helmet assembly of clause I, wherein said bottom member of each of said first and said second posts comprises a second joint defining a third pivot to allow said lower beam to rotate about said third pivot defined by said second joint.

(163) IX. A surgical helmet assembly for use with a surgical garment including a coupling element, said assembly comprising: a surgical helmet comprising: a headband for securing said surgical helmet to a wearer; a ventilation unit supported by the headband; a face frame slidably coupled to said surgical helmet, said face frame comprising: an upper beam comprising opposing regions; a first post and a second post, each having a first end and a second end, said first end of each of said first and said second posts being coupled to said opposing regions of said upper beam; a lower beam connecting said second ends of said first and said second posts; wherein each of said first post and said second post comprises a top member and a bottom member, said bottom member being movably coupled to said top member; and wherein each of said first post and said second post are movable between a first position defining a first configuration of said face frame and a second position defining a second configuration of said face frame, each of said configurations of said face frame having a different distance between said upper beam and said lower beam.

(164) X. The surgical helmet assembly of clause IX, further comprising: a first surgical garment comprising: a first shield having an upper portion and an opposing lower portion; a first aperture in said upper portion of said first shield; and a first coupling element disposed on said lower portion of said first shield; wherein said first aperture and said first coupling element define a first coupling dimension; a second surgical garment comprising: a second shield having an upper portion and an opposing lower portion; a second aperture in said upper portion of said second shield; and a second coupling element disposed on said lower portion of said second shield; wherein said second aperture and said second coupling element define a second coupling dimension; wherein said first coupling dimension of said first surgical garment corresponds to the first configuration of said face frame to allow said first surgical garment to couple to said face frame when said first post and said second post are in said first position; and wherein said second coupling dimension of said second

surgical garment corresponds to the second configuration of said face frame to allow said second surgical garment to couple to said face frame when said first post and said second post are in said second position.

(165) XI. A surgical helmet assembly for use with a surgical garment including a transparent shield, said assembly comprising: a surgical helmet comprising a coupling feature disposed on a distal portion of said surgical helmet, said coupling feature comprising a plurality of retention features spaced about a distal surface of said coupling feature; a face frame removably coupled to said surgical helmet, said face frame comprising: a coupling mechanism configured to operatively engage said coupling feature of said surgical helmet; and wherein said coupling mechanism and said coupling feature comprise complementary features to allow adjustment of the position of said face frame relative to said surgical helmet to position the transparent shield of the surgical garment at a preferred position in front of the wearer's face when disposed over said surgical helmet.

(166) XII. The surgical helmet assembly of clause XI, wherein said coupling feature of said surgical helmet further comprises a channel; and wherein said coupling mechanism of said face frame comprises a coupling arm including a complementary protrusion slidably disposed within said channel of said coupling feature when said face frame is coupled to said surgical helmet.

(167) XIII. The surgical helmet assembly of clause XI, wherein said coupling mechanism further comprises a control member configured to selectively manipulate a first member between a first state and a second state; wherein when said first member is in said first state, said coupling mechanism may move freely relative to said coupling feature of said surgical helmet; and wherein when said first member is in said second state, said coupling mechanism is locked in the desired position relative to said coupling feature of said surgical helmet.

(168) XIV. The surgical helmet assembly of clause XI, wherein said face frame further comprises: an upper beam including opposing regions; a first post and a second post, each having a first end and a second end, said first end of each of said first and said second posts being coupled to said opposing regions of said upper beam; a lower beam connecting said second ends of said first and said second posts; and wherein each of said first post and said second post comprises a top member and a bottom member, said bottom member being movably coupled to said top member to provide a variable length of each of said first post and said second post to vary the distance between said upper beam and said lower beam.

(169) XV. The surgical helmet assembly of clause XI, wherein said face frame further comprises: an upper beam and an opposing lower beam that define a first coupling dimension; and said surgical helmet assembly further comprises a second face frame that is removably coupled to said surgical helmet, said second face frame comprising: a second upper beam and an opposing second lower beam that define a second coupling dimension; wherein said first coupling dimension of said face frame is larger than said second coupling dimension of said second face frame such that said first face frame is configured to fit a wearer with a larger head and said second face frame is configured to fit a wearer with a smaller head.

(170) XVI. A surgical helmet for use with a surgical garment including a coupling element, said helmet comprising: a headband; a ventilation unit supported by the headband; a face frame comprising: an upper beam comprising a first region and an opposing second region; a first post and a second post, each having a first end and an opposing second end, each of said first and said second posts coupled to said upper beam; a lower beam connecting said second ends of said first and said second posts; wherein opposing ends of said lower beam are connected to said second ends of said first and said second posts by a first joint configured to allow the lower beam to rotate relative to said first and said second posts.

(171) XVII. The surgical helmet of clause XVI, wherein said upper beam comprises a coupling member disposed on each of said first region and said second region of said upper beam for coupling said first and said second posts to said upper beam; wherein said coupling member defines a second joint between said upper beam and said first and said second posts, said second

joint configured to allow each of said first and said second posts to rotate relative to said upper beam.

(172) XVIII. The surgical helmet of clause XVII, wherein said coupling member is configured to removably engage said first and said second posts.

(173) XIX. The surgical helmet of clause XVI, further comprising an attachment element that is at least partially disposed within said lower beam, such that said attachment element may be repositioned relative to said upper beam when said lower beam is rotated relative to said first and said second posts.

(174) XX. A surgical apparel system comprising: a surgical helmet comprising a mounting feature disposed on a distal portion of said surgical helmet; a surgical garment configured to be at least partially disposed over said surgical helmet, said surgical garment comprising: a fabric defining an opening; and a transparent shield disposed within said opening, said transparent shield comprising: an upper portion and a lower portion; a left portion and a right portion separated by a medial axis that bisects said transparent shield; and a left edge and opposing right edge; an attachment element coupled to said upper portion of said transparent shield and configured to removably couple with said mounting feature of the surgical helmet; and an adjustment device disposed on said lower portion of said transparent shield, said adjustment device comprising a first end and a second end; wherein said adjustment device is configured to expand and contract to vary a distance between said first end and said second end; and wherein said first end and said second end are spaced laterally across said lower portion of said transparent shield such that said expansion or contraction of said adjustment device to vary that distance between said first end and said second end will manipulate a curvature of said transparent shield to increase a user's field of view when said surgical garment is at least partially disposed over said surgical helmet.

(175) XXI. The surgical apparel system of clause XX, wherein said adjustment device comprises a cylinder including a piston configured to move longitudinally within said cylinder manipulating the distance between said first end and said second end of the adjustment device to vary the curvature of said lower portion of said transparent shield.

(176) XXII. The surgical apparel system of clause XX, wherein said adjustment device comprises a ratchet coupled to said transparent shield; and a strap having a first end and a second end, wherein said first end is coupled to said transparent shield at position laterally spaced from said ratchet; and wherein said ratchet and said strap are configured such that said second end of said strap is fed through said ratchet to reduce the distance between said ratchet and said first end of said strap to vary the curvature of said lower portion of said transparent shield.

(177) XXIII The surgical apparel system of clause XX, wherein said transparent shield further comprises a tab extending from the upper portion of said transparent shield; and an opening defined by said tab, said opening configured to removably couple said surgical garment with the surgical helmet.

(178) XXIV. The surgical apparel system of clause XXIII, wherein said surgical helmet further comprises an upper beam, said upper beam defining an alignment channel; and a protrusion coupled to said upper beam and at least partially disposed within said alignment channel; and wherein said tab is configured to be at least partially disposed within said alignment channel when said protrusion is inserted within said opening to couple said surgical garment to said surgical helmet.

(179) XXV. A surgical helmet assembly comprising: a surgical helmet comprising: a headband; a ventilation unit supported by the headband; a first face frame configured to be removably coupled to said surgical helmet, said first face frame comprising: a first upper beam comprising opposing regions and a first coupling feature disposed between said opposing regions; a first pair of posts that each comprise a first end and a second end, said first end of each of said first pair of posts being coupled to said opposing ends of said first upper beam; a first lower beam connecting said second ends of said first pair of posts; and a first attachment element at least partially disposed

within said first face frame, said first coupling feature and said first attachment element defining a first coupling dimension; a second face frame configured to be removably coupled to said surgical helmet, said second face frame comprising: a second upper beam comprising opposing regions and a second coupling feature disposed between said opposing regions; a second pair of posts that each comprise a first end and a second end, said first end of each of said second pair of posts being coupled to said opposing ends of said second upper beam; a second lower beam connecting said second ends of said second pair of posts; and a second attachment element at least partially disposed within said second face frame, said second coupling feature and said second attachment element defining a second coupling dimension; a surgical garment comprising: a shield having an upper portion and an opposing lower portion; an aperture in said upper portion of said shield; a first coupling element disposed on said lower portion of said shield; and a second coupling element disposed on said lower portion of said shield; wherein a distance between said aperture and said first coupling element define a third coupling dimension; and wherein a distance between said aperture and said second coupling element define a fourth coupling dimension; wherein said first coupling dimension of said first face frame corresponds to said third coupling dimension of said surgical garment to allow said surgical garment to couple to said first face frame; and wherein said second coupling dimension of said second face frame corresponds to said fourth coupling dimension of said surgical garment to allow said surgical garment to couple to said second face frame.

(180) XXVI. A surgical helmet assembly comprising: a surgical helmet comprising: a headband for securing said helmet to a wearer; a ventilation unit supported by the headband; a first face frame configured to be removably coupled to said surgical helmet, said first face frame comprising: a first upper beam comprising opposing regions and a first coupling feature disposed between said opposing regions; a first pair of posts that each comprise a first end and a second end, said first end of each of said first pair of posts being coupled to said opposing ends of said first upper beam; a first lower beam connecting said second ends of said first pair of posts; and a first attachment element at least partially disposed within said first face frame, said first coupling feature and said first attachment element defining a first coupling dimension; a second face frame configured to be removably coupled to said surgical helmet, said second face frame comprising: a second upper beam comprising opposing regions and a second coupling feature disposed between said opposing regions; a second pair of posts that each comprise a first end and a second end, said first end of each of said second pair of posts being coupled to said opposing ends of said second upper beam; a second lower beam connecting said second ends of said second pair of posts; and a second attachment element at least partially disposed within said second face frame, said second coupling feature and said second attachment element defining a second coupling dimension; a first surgical garment comprising: a first shield having an upper portion and an opposing lower portion; a first aperture in said upper portion of said first shield; and a first coupling element disposed on said lower portion of said first shield; wherein said first aperture and said first coupling element define a third coupling dimension; a second surgical garment comprising: a second shield having an upper portion and an opposing lower portion; a second aperture in said upper portion of said second shield; and a second coupling element disposed on said lower portion of said second shield; wherein said second aperture and said second coupling element define a fourth coupling dimension; wherein said first coupling dimension of said first face frame corresponds to said third coupling dimension of said first surgical garment to allow said first surgical garment to couple to said first face frame; and wherein said second coupling dimension of said second face frame corresponds to said fourth coupling dimension of said second surgical garment to allow said second surgical garment to couple to said second face frame.

(181) XXVII. A surgical apparel system comprising: a surgical helmet comprising a mounting feature disposed on a distal portion of said surgical helmet; a face frame removably coupled to said surgical helmet, said face frame comprising: a first post and a second post laterally spaced from

one another, each of said first and second posts comprising a first end and a second end; a lower beam extending between the second end of each of said first and second posts; a first upper beam coupled to and extending proximally from said first end of said first post; a second upper beam coupled to and extending proximally from said first end of said second post; a face frame coupler configured to removably engage said mounting feature of said surgical helmet; and a coupling member comprising one of a ferromagnetic material or a magnetic material; a first surgical garment comprising: a fabric defining an opening; a transparent shield disposed within said opening; and an attachment member element comprising the other of the ferromagnetic material or the magnetic material, said attachment element configured to removably engage said coupling member to couple said first surgical garment to said face frame; a second surgical garment comprising: a fabric defining an opening; and a transparent shield disposed within said opening, said transparent shield comprising: an upper portion and a lower portion; an attachment element coupled to said upper portion of said transparent shield and configured to removably couple with said mounting feature of the surgical helmet; and an adjustment device disposed on said lower portion of said transparent shield configured to manipulate the a curvature of said transparent shield; wherein said system comprises a first configuration comprising said first surgical garment being at least partially disposed over said surgical helmet; and wherein said system comprises a second configuration comprising said second surgical garment being at least partially disposed over said surgical helmet.

(182) XXVIII. A face frame for use with a surgical helmet including a mounting feature, said face frame comprising: a first post and a second post laterally spaced from one another, each of said first and second posts comprising a first end and a second end; a lower beam extending between the second end of each of said first and second posts; a first upper bracket coupled to and extending proximally from said first end of said first post; a second upper bracket coupled to and extending proximally from said first end of said second post; and a counterweight disposed on each of said first and second upper brackets positioned proximal to said first and second posts; and wherein said counterweight are positioned on each of said first and second upper brackets a distance from said first and second posts sufficient to maintain sagittal position of a center of gravity of the surgical helmet when said face frame is coupled to the surgical helmet.

(183) XXIX. The face frame of clause XXVIII, wherein said face frame further comprises an upper beam; wherein said upper beam comprises a face frame coupler comprising one of a ferromagnetic material or a magnetic material; and wherein the mounting feature of the surgical helmet comprises the of the ferromagnetic material or the magnetic material for removably coupling said face frame to the surgical helmet.

(184) XXX. A surgical helmet assembly for use with a surgical garment including a transparent shield, said assembly comprising: a surgical helmet comprising: a mounting feature disposed on a distal portion of said surgical helmet; a face frame removably coupled to said surgical helmet, said face frame comprising: a first post and a second post laterally spaced from one another, each of said first and second posts comprising a first end and a second end; a lower beam extending between the second end of each of said first and second posts; a first upper bracket coupled to and extending proximally from said first end of said first post; a second upper bracket coupled to and extending proximally from said first end of said second post; and a face frame coupler configured to operatively engage said mounting feature of said surgical helmet.

(185) XXXI. The assembly of clause XXX, wherein said face frame coupler and said mounting feature comprise complementary features to allow adjustment of the position of said face frame relative to said surgical helmet to position the transparent shield of the surgical garment at a preferred position in front of a wearer's face when disposed over said surgical helmet.

(186) XXXII. The assembly of clause XXX, further comprising a counterweight disposed on each of said first and second upper brackets proximal to said first and second posts.

(187) XXXIII The assembly of clause XXXII, wherein said counterweight comprises a mass and is positioned on each of said first and second upper brackets a distance from said first and second

posts sufficient to maintain sagittal position of the center of gravity of the surgical helmet when face frame is coupled to the surgical helmet.

(188) XXXIV. The assembly of clause XXX, further comprising an upper beam extending between the first end of each of said first and second posts; wherein said upper beam defines an alignment channel; and wherein said upper beam further comprises a protrusion at least partially disposed in said alignment channel.

(189) XXXV. The assembly of clause XXX, wherein said face frame coupler is positioned proximate the intersection of said first and second upper brackets with said first and second posts.

(190) XXXVI. The assembly of clause XXX, wherein said face frame coupler comprises an aperture defined in each of said first upper bracket and said second upper bracket; and wherein said mounting feature of said surgical helmet comprises a first post and a second post, each of said first post and said second post configured to be inserted through one of said apertures in said first upper bracket and said second upper bracket.

(191) XXXVII. The assembly of clause XXX, wherein said face frame coupler comprises one of a magnet or a ferromagnetic material first upper bracket and said second upper bracket; and wherein said mounting feature of said surgical helmet comprises the other of said magnet or said ferromagnetic material configured to magnetically couple said face frame to said helmet.

(192) XXXVIII. The assembly of clause XXX, wherein said surgical helmet further comprises an upper beam coupled to said distal portion of said surgical helmet, said upper beam comprising opposing lateral portions; wherein said mounting feature of said surgical helmet comprises a post disposed on each of said opposing lateral portions; and wherein said face frame coupler comprises an aperture defined in each of said first upper bracket and said second upper bracket configured to removably couple with one of said posts on said opposing lateral portions to couple said face frame to said surgical helmet.

(193) XXXIX. The assembly of clause XXX, wherein said surgical helmet further comprises an upper beam coupled to said distal portion of said surgical helmet, said upper beam comprising opposing lateral portions; wherein said mounting feature of said surgical helmet comprises one of a magnet or a ferromagnetic material disposed on and/or within each of said opposing lateral portions; and wherein said face frame coupler comprises the other of said magnet or said ferromagnetic material disposed on and/or within each of said first upper bracket and said second upper bracket configured to magnetically couple said face frame to said helmet.

(194) XL. A surgical apparel system comprising: a surgical helmet comprising a mounting feature disposed on a distal portion of said surgical helmet; a face frame removably coupled to said surgical helmet, said face frame comprising: a first post and a second post laterally spaced from one another, each of said first and second posts comprising a first end and a second end; a lower beam extending between the second end of each of said first and second posts; a first upper beam coupled to and extending proximally from said first end of said first post; a second upper beam coupled to and extending proximally from said first end of said second post; a face frame coupler configured to removably engage said mounting feature of said surgical helmet; and a coupling member comprising one of a ferromagnetic material or a magnetic material; a first surgical garment comprising: a fabric defining an opening; a transparent shield disposed within said opening; and an attachment member element comprising the other of the ferromagnetic material or the magnetic material, said attachment element configured to removably engage said coupling member to couple said first surgical garment to said face frame; a second surgical garment comprising: a fabric defining an opening; and a transparent shield disposed within said opening, said transparent shield comprising: an upper portion and a lower portion; an attachment element coupled to said upper portion of said transparent shield and configured to removably couple with said mounting feature of the surgical helmet; and an adjustment device disposed on said lower portion of said transparent shield configured to manipulate the a curvature of said transparent shield; wherein said system comprises a first configuration comprising said first surgical garment being at least partially

disposed over said surgical helmet; and wherein said system comprises a second configuration comprising said second surgical garment being at least partially disposed over said surgical helmet. (195) XLI. A face frame for use with a surgical helmet including a mounting feature, said face frame comprising: a first post and a second post laterally spaced from one another, each of said first and second posts comprising a first end and a second end; a lower beam extending between the second end of each of said first and second posts; a first upper bracket coupled to and extending proximally from said first end of said first post; a second upper bracket coupled to and extending proximally from said first end of said second post; and a face frame coupler positioned proximate the intersection of said first and second upper brackets with said first and second posts, said face frame coupler configured to removably engage the mounting feature of the surgical helmet to allow for removal of said face frame from the surgical helmet to reduce the weight of the surgical helmet when said face frame is not needed.

(196) Several configurations have been discussed in the foregoing description. However, the configurations discussed herein are not intended to be exhaustive or limit the system **10, 110** and/or the surgical helmet **20, 120, 220, 320, 420** to any particular form. The terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations are possible in light of the above teachings and the system **10, 110** may be practiced otherwise than as specifically described. Furthermore, it should be understood that elements described in the various configurations including reference numbers in increments of 100 may comprise similar features.

Claims

1. A face frame for use with a surgical helmet including a mounting feature comprising one of a ferromagnetic material or a magnetic material, said face frame comprising: a first post and a second post laterally spaced from one another, each of said first and second posts comprising an upper portion and a lower portion; a lower beam extending between the lower portion of each of said first and second posts; and a face frame coupler positioned proximate a first end of each of said first and second posts, said face frame coupler comprising either of the ferromagnetic material or a magnetic material that is capable of removably engaging the mounting feature of the surgical helmet to allow for removal of said face frame from the surgical helmet to reduce a weight of the surgical helmet when said face frame is not needed; an upper beam extending between the upper portion of each of said first and second posts; and a protrusion that is removably coupled to said upper beam to allow for said protrusion to be repositioned on said upper beam based on the length of said first post and said second post.
2. The face frame of claim 1, further comprising: the upper beam coupled to and extending proximally from said upper portion of each of said first post and said second post; and wherein said face frame coupler is positioned proximate an intersection of said upper beam with said first and second posts.
3. The face frame of claim 2, further comprising the upper beam extending between the upper portion of each of said first and second posts; wherein said upper beam defines an alignment channel; and wherein said upper beam further comprises a protrusion at least partially disposed in said alignment channel.
4. The face frame of claim 2, further comprising a counterweight disposed on a bracket coupled to said upper beam proximal to said first and second posts.
5. The face frame of claim 4, wherein said counterweight comprises a mass and is positioned on the first and second lateral portions of said upper beam a distance from said first and second posts sufficient to maintain sagittal position of the center of gravity of the surgical helmet when said face frame is coupled to the surgical helmet.
6. The face frame of claim 4, wherein said counterweight and each of said first and second lateral

portions of said upper beam comprise complementary fasteners, said fasteners configured to allow said counterweight may be positioned at various points spaced along a length of said first and second lateral portions of said upper beam to manipulate the position of the center of gravity of the surgical helmet when said face frame is coupled to the surgical helmet.

7. The face frame of claim 1, further comprising a coupling member disposed on at least one of said first post or said second post, said coupling member comprising one of a ferromagnetic material or a magnetic material.

8. The face frame of claim 1, wherein each of said first post and said second post further comprises a top member and a bottom member, said bottom member being movably coupled to said top member to provide a variable length of each of said first post and said second post to vary a distance between said lower beam and the surgical helmet.

9. The face frame of claim 1, wherein each of said first post and said second post further comprises a top member and a bottom member, said bottom member being movably coupled to said top member to provide a variable length of each of said first post and said second post to vary a distance between said lower beam and the surgical helmet.
